NILAMBARA ELECTRICALS

# System Testing and Test Results

Repair Shop Management System

Version 1.0 | Generated 21 October 2026

Fresh isolated execution and BRD traceability record

482/A/3, Colombo Road, Kaduwela, Sri Lanka.

Currency: LKR | Timezone: Asia/Colombo

Based on the existing implementation, the original BRD and isolated test evidence. Business-owner acceptance remains pending.

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# 1. Document control

| Document control | Value |
|---|---|
| Version / generation | 1.0 / 21 October 2026 |
| Scope | Actual Step 13 execution, source inspection and manual acceptance boundaries |
| Basis | Original BRD (20 pages, 27 sections); source/route/schema inspection; fresh Step 13 JUnit and native checks |
| Author / approval | Codex documentation runner / business-owner approval pending |
| Verification boundary | Automated scenario evidence and real fixture screenshots; no production deployment or live business-data writes |
| Companion sources | documentation/sources/; documentation/evidence/; documentation/screenshots/ |

# 2. Testing objectives

Verify implemented operations and integration relationships against the original BRD; preserve exact test evidence, identify gaps and produce a technical assessment without modifying genuine business records.

# 3. Scope

All existing Unit and Feature tests were executed, followed by native POS/invoice/payment/receipt/technician/repair-stock/GRN races and real disposable backup recovery. Source inspection covered routes, controllers, services, models, migrations, Blade, Livewire, config, all BRD sections and current MySQL metadata. Documentation and test execution only: no hosting/build/deployment, schema redesign or business logic changes.

| Inspection inventory | Count |
|---|---|
| app | 163 |
| routes | 19 |
| database | 30 |
| resources | 84 |
| tests | 27 |
| config | 17 |

# 4. Environment

| Component | Actual finding |
|---|---|
| PHP | 8.5.11 |
| Laravel | 13.37.0 |
| Database connection/server | mysql / 10.4.32-MariaDB |
| Business database | nilambara_repair_management (read-only inspection only) |
| Test database | Fresh randomized nilambara_step2_test_<12 hex>; created/dropped by guarded helper |
| Browser | Installed local Chrome headless via DevTools; real isolated Laravel fixture app on loopback 8087 |
| UI assets | Existing built assets reused; no production build |
| Schema | 69 tables; 199 FK constraints; all InnoDB/utf8mb4 |
| Timezone / currency | Asia/Colombo / LKR |
| Execution date | 2026-10-20 |

# 5. Methodology

scripts/verify-step13.php is a documentation-only copy of the existing guarded verifier, adding JUnit output. It creates a nonexistent random database, passes explicit testing DB environment to children, runs the full suite, rolls back/remigrates only that disposable database, performs native multi-process/recovery checks, then drops only its own database. Before/after operational metadata/count snapshots were compared. A separate guarded database and non-sensitive fixtures powered screenshots; that database was removed after capture. No operational web login or transaction was used for screenshots.

Per-case records map JUnit method/data-set names to executable test source line, request/service sequence and assertion operations. Expected behavior is the objective encoded in those assertions. The original source and raw logs remain authoritative for exact fixture literals and exception branches; no console output is fabricated.

# 6. Requirement traceability

The original BRD contains 27 numbered sections on 20 pages. The maintained matrix contains 327 requirement rows. Suite-level/field-only mappings are retained with their limits; the status of a BRD requirement is separate from a passing test method. Appendix 24 includes every matrix row with original requirement ID.

| Matrix status | Rows |
|---|---|
| Partially Implemented | 54 |
| Implemented and Verified | 209 |
| Implemented but Not Tested | 39 |

| Matrix status | Rows |
|---|---|
| Not Applicable | 4 |
| Blocked | 21 |

# 7. Module test coverage

| Documented module area | Passed executed cases (primary classification) |
|---|---|
| Authentication | 11 |
| Business dashboard | 6 |
| Customer and company management | 14 |
| Repair receipt management | 13 |
| Repair job management | 0 |
| Quotation management and decisions | 11 |
| Invoice management | 16 |
| Payment management | 10 |
| Credit management | 0 |
| Stock and inventory | 37 |
| Supplier management | 0 |
| Purchase and GRN management | 14 |
| Point of sale | 17 |
| Reports and analysis | 10 |
| Employee and account administration | 19 |
| Roles and permission controls | 0 |
| System settings | 3 |
| Print template management | 4 |
| Audit history | 25 |
| Backup and restore | 8 |

Each test is classified once in this table. Cross-module assertions cover additional areas; a zero primary classification is not a claim that a module is absent. The per-case source/method and feature catalogue supply granular scenario links.

# 8. Integration testing

The actual intake-to-delivery service scenario receives an advance, issues/approves a quotation, issues parts, completes the job, invoices and collects the remainder, then records delivery. It verifies the approved total without manually forcing job selling prices. Purchasing tests cover draft/finalize/partial GRN/payment/reversal and return credits. Existing multi-job, payment/credit and POS scenarios verify their linked tables and protected replay. Browser capture proves page availability/rendering, not completion of those transactions through click automation.

# 9. Database relationships

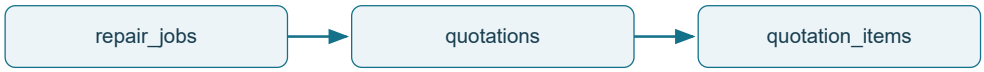
| Audit check | Actual result |
|---|---|
| tables | 69 |
| foreign_key_constraints | 199 |
| orphans | 0 |
| all_innodb | True |
| all_utf8mb4 | True |
| floating_columns | [] |
| row_counts_unchanged | True |

Primary/unique/index metadata and FK columns/delete rules are preserved in schema-before/after.json. Full composite constraints were checked as tuples, in addition to per-column mappings. No orphan found. BusinessConstraintsTest/SchemaIntegrityTest/BusinessRelationshipsTest execute required-field, uniqueness, restrict/cascade and cross-customer/target integrity behavior; model method names alone are not evidence. Matching row counts are not a byte-for-byte operational record comparison.

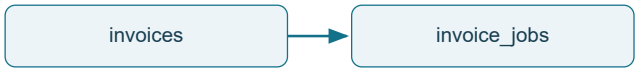
# Repair and financial relationships



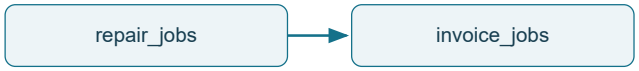

Parent to referencing child; arrows correspond to inspected foreign keys.



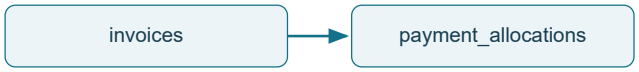

Parent to referencing child; arrows correspond to inspected foreign keys.



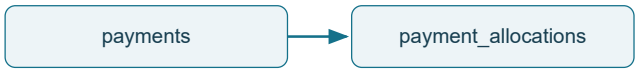

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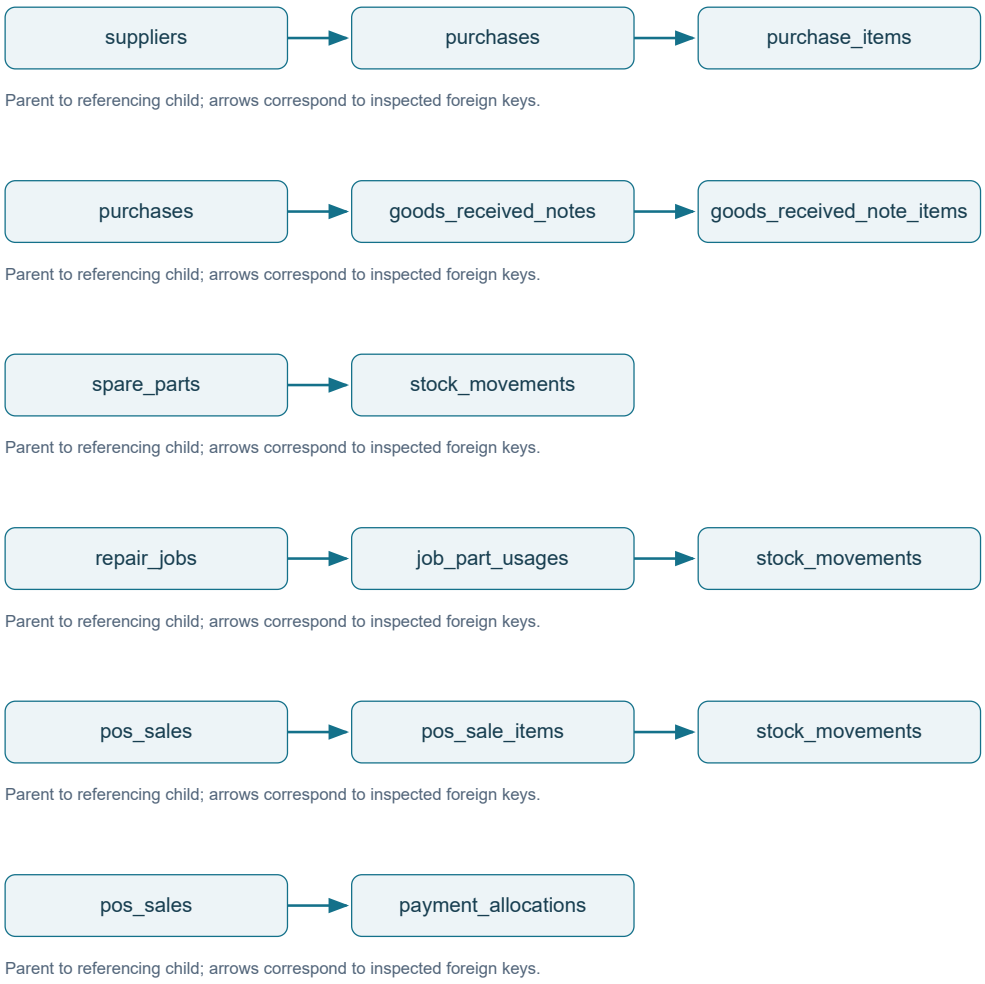

Parent to referencing child; arrows correspond to inspected foreign keys.




Parent to referencing child; arrows correspond to inspected foreign keys.

Verified table relationships, not a sequence of automatic business actions. Join tables connect the separate paths.

### Shared stock, purchase and POS relationships




Verified table relationships, not a sequence of automatic business actions. Join tables connect the separate paths.

# 10. Financial calculations

Fresh suites assert exact decimals for invoice totals, advances/allocations, outstanding/partial states, credit collections, refunds/reversals, supplier openings/settlements and latest-purchase profit/valuation. The existing integrated fixture verifies a 30.00 invoice, 5.00 advance and 25.00 remainder; known cost/labour gives 22.00 recorded-cost estimate. These are isolated test assertions, not operational invoice values. Payment accounting reversal affects collections without removing issued sales. Supplier targets reject overpayment. Allocation is not a second receipt. Purchase-wide adjustment allocation remains undecided; affected estimates remain incomplete.

# 11. Inventory testing

Opening movement and known cost, partial GRNs, request-without-deduction, exactly-once issue, full unused resalable issue return, correction/damage/count, supplier return, POS deduction/return and shared concurrent access are covered. Native repair issue versus POS and two-storekeeper GRN races verify common stock/remaining-receipt constraints. Unit

precision and negative availability are validated. Partial single-issue returns and ambiguous historical repeated-part source returns remain implementation boundaries, not failed claims hidden by tests.

# 12. Security testing

Executed authentication scenarios cover required credentials, rejection, throttling, CSRF with verification active, session regeneration/logout invalidation, disabled accounts and disabled public routes. Role tests cover direct access, explicit deny, last administrator and assigned-technician restrictions. Targeted SQL-looking input and stored script escaping probes ran. Audit secret redaction, setting allowlists, backup private-root/traversal/checksum protections and authenticated restore safeguards ran. This is not a comprehensive penetration test or certification of production HTTPS/Windows filesystem controls. No authentication/CSRF safeguard was disabled to claim a passing result.

# 13. Roles and permissions

The following grants were read from the operational role relations, not inferred from the default seeder. Individual override count at inspection: 0. Subsequent administrator customization can change access. Some actions need additional record policy or several permissions together.

## Administrator

Manage accounts/roles/settings/templates/audit and protected backups. Preserve the final active administrator. Restore is CLI-only and requires explicit target approval.

Current permission grants (98): audit.view, backups.configure, backups.create, backups.download, backups.restore, backups.view, credit.collect, credit.reports, credit.view, customers.create, customers.deactivate, customers.statement, customers.update, customers.view, dashboard.view, employees.create, employees.deactivate, employees.update, employees.view, inventory.adjust, inventory.create, inventory.issue, inventory.stock-count, inventory.update, inventory.view, invoices.cancel, invoices.create, invoices.export, invoices.issue, invoices.print, invoices.update-draft, invoices.view, job-cards.print, job-cards.view, jobs.assign, jobs.cancel, jobs.complete, jobs.create, jobs.inspect, jobs.repair, jobs.update, jobs.update-status, jobs.view, payments.create, payments.export, payments.print, payments.receive, payments.receive-for-others, payments.reverse, payments.view, permissions.manage, pos.discount, pos.payment.verify, pos.receipt.print, pos.refund, pos.reports.view, pos.return, pos.sale, pos.sale.cancel, pos.sales.view, pos.view, print-templates.reset, print-templates.update, print-templates.view, purchases.approve, purchases.create, purchases.receive, purchases.view, quotations.approve, quotations.cancel, quotations.create, quotations.issue, quotations.print, quotations.revise, quotations.update, quotations.view, receipts.create, receipts.print, receipts.reprint, receipts.view, reports.credit, reports.export, reports.financial, reports.inventory, reports.repair, reports.sales, reports.view, roles.manage, roles.view, settings.update, settings.view, suppliers.create, suppliers.update, suppliers.view, users.create, users.disable, users.update, users.view

## Receptionist

Search/register customers; receive repairs and print intake/card documents; prepare quotations where granted. No assumed authority for money, stock, financial statements or administration.

Current permission grants (19): customers.create, customers.update, customers.view, dashboard.view, invoices.view, job-cards.print, job-cards.view, jobs.create, jobs.view, quotations.create, quotations.issue, quotations.print, quotations.update, quotations.view, receipts.create, receipts.print, receipts.view, reports.repair, reports.view

## Technician

Inspect and repair assigned jobs, update their status and request parts. Cannot use another technician's job by changing its URL. Stock issue and customer-money authority are separate.

Current permission grants (10): dashboard.view, job-cards.view, jobs.complete, jobs.inspect, jobs.repair, jobs.update-status, jobs.view, quotations.view, reports.repair, reports.view

### Cashier

Create/issue invoices, receive actual collections and advances, collect credit and perform permitted POS sales/printing. No default refund/reversal, supplier or settings authority.

Current permission grants (23): credit.collect, credit.view, customers.view, dashboard.view, invoices.create, invoices.export, invoices.issue, invoices.print, invoices.update-draft, invoices.view, payments.create, payments.export, payments.print, payments.receive, payments.view, pos.receipt.print, pos.sale, pos.sales.view, pos.view, quotations.print, quotations.view, reports.sales, reports.view

### Storekeeper

Maintain parts, request/issue/adjust/count stock and enter/receive purchases. Purchase approval, supplier administration and supplier money are management-only under current grants.

Current permission grants (14): dashboard.view, inventory.adjust, inventory.create, inventory.issue, inventory.stock-count, inventory.update, inventory.view, pos.view, purchases.create, purchases.receive, purchases.view, reports.export, reports.inventory, reports.view

### Manager

Assign repairs, prepare/approve quotations, supervise invoices/collections/credit, suppliers/purchase approval and financial reports. No assumed settings/account/restore access.

Current permission grants (52): credit.collect, credit.reports, credit.view, customers.statement, customers.view, dashboard.view, invoices.cancel, invoices.create, invoices.export, invoices.issue, invoices.print, invoices.update-draft, invoices.view, jobs.assign, jobs.view, payments.create, payments.export, payments.print, payments.receive, payments.reverse, payments.view, pos.discount, pos.payment.verify, pos.receipt.print, pos.refund, pos.reports.view, pos.return, pos.sale.cancel, pos.sales.view, pos.view, purchases.approve, purchases.create, purchases.receive, purchases.view, quotations.approve, quotations.cancel, quotations.create, quotations.issue, quotations.print, quotations.revise, quotations.update, quotations.view, reports.credit, reports.export, reports.financial, reports.inventory, reports.repair, reports.sales, reports.view, suppliers.create, suppliers.update, suppliers.view

Operational grants were inspected read-only; no role seeder was run on the operational database. Fresh fixtures use seeded grants and therefore may differ from existing customized grants. Authorization includes route middleware, controller/service checks and record policy; a visible link is not the enforcement boundary.

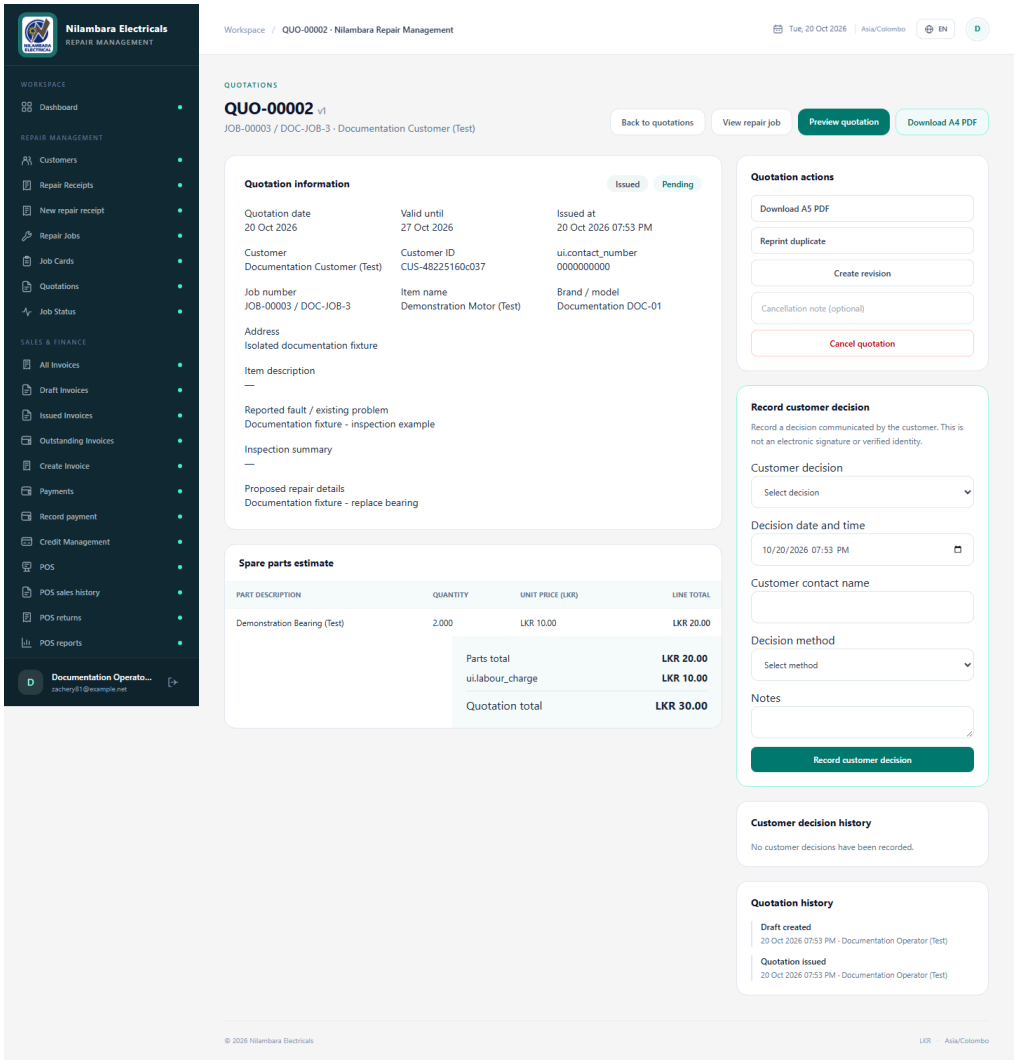
# 14. Printing and Unicode

Existing automated scenarios exercise receipt/card, quotation, invoice, payment, POS, statements, purchase/GRN and report outputs with permissions, PDF signatures, templates/duplicate histories and local fonts. These three documentation PDFs are rendered by actual Chrome with local Arial and Noto Sans Sinhala and are visually reviewed after generation. Physical A5/A6/A4/thermal printing, duplex/real hardware and scanner tests remain Not Run. Screenshot render of an HTML print view does not prove a physical device printed.

The following existing Sinhala UI labels are copied directly from lang/si/ui.php, not newly translated:

පාලන පුවරුව | පාරිභෝගිකයන්

Some actual application quotation labels display unresolved translation keys (for example ui.contact_number and ui.labour_charge); see D13-01. This task records the issue and does not alter business logic/translations.




Actual isolated issued quotation and customer-decision page. Unresolved labels are recorded as D13-01; fixture amounts are not operational transactions.

# 15. Backup and restore

Native checks created a nonempty real test mysqldump, verified SHA-256/private protection, rejected wrong-password/confirmation, created safety backup, restored fixture data through authenticated CLI, checked maintenance lifecycle and recovery audit history. Isolated runtime prevents operational maintenance/restore effects. Default policy inspected: 02:00 Asia/Colombo, 30 days, at most 30 scheduled copies. No real Drive upload or operational destination acceptance is claimed; those are Blocked. Host scheduler was not activated.

# 16. Performance and concurrency

- PASS: two cashiers buying four of five units; one sale, one rejection, one unit remains.
- PASS: simultaneous independent checkouts receive distinct sale numbers.
- PASS: simultaneous identical retries create one sale and one payment.
- PASS: two simultaneous cashiers, invoice, one save and one rejection; barrier-to-completion 165.11 ms.
- PASS: two simultaneous cashiers, payment, one save and one rejection; barrier-to-completion 248.532 ms.

- PASS: one invoice-job reservation and exactly 100.00 allocated, no overpayment.
- PASS: service/PDF observations recorded: {"invoice-create-and-issue":60.935,"payment-receive-and-allocate":77.373,"pos-checkout-including-stock-posting":41.393,"a5-invoice-pdf-with-print-history":1419.177}
- PASS: two receptionists create atomic receipts and unique jobs; 2 concurrent workers; 250.599 ms; ["SAVED:1","SAVED:2"]
- PASS: two technicians update their assigned jobs with retained histories; 2 concurrent workers; 183.713 ms; ["SAVED:5","SAVED:6"]
- PASS: POS and repair issue compete for the same five units; 2 concurrent workers; 300.059 ms; ["SAVED:1","REJECTED:{\"cart\":[\"Insufficient saleable stock for Test spare part. Available: 1.000.\"]}"]
- PASS: two storekeepers cannot over-receive a purchase; 2 concurrent workers; 183.369 ms; ["SAVED:1","REJECTED:{\"quantity\":[\"Quantity exceeds the unreceived purchase quantity.\"]}"]
- PASS: receipt, technician, shared stock and GRN race invariants verified in disposable MySQL.
- PASS: real MySQL dump created, nonempty, SHA-256 verified, private and protected.
- PASS: confirmation and wrong-password refusals; authenticated CLI restore, safety dump, maintenance lifecycle, data recovery and audit history.

Native lines are recorded as supplementary invariant checks, not added to PHPUnit totals. The timing JSON and raw log contain exact observations; single small-data local timings are not throughput or realistic-volume capacity acceptance.

# 17. Results summary

| Register | Total | Passed | Failed | Blocked | Not Run |
|---|---|---|---|---|---|
| Automated JUnit | 218 | 218 | 0 | 0 | 0 |
| Manual/external acceptance | 12 | 0 | 0 | 4 | 8 |
| Combined declared register | 230 | 218 | 0 | 4 | 8 |

Executed automated pass rate: 100.00%. Assertions: 3566. JUnit summed time: 171.71s; console command duration is preserved in step13-tests.log. This percentage applies only to executed tests, not to all BRD features or business acceptance.

# 18. Failed tests and defects

Fresh execution had no failed/error test cases. The historical defect register is preserved, including fixed items and remaining external/functional boundaries. No application fix was made during Step 13. Open counts below include declared limitations/configuration dependencies; they are not newly failed tests.

| Open severity | Count |
|---|---|
| Critical | 0 |
| High | 1 |

| Open severity | Count |
|---|---|
| Medium | 2 |
| Low | 2 |

## D12-01 - Discounted refund cent rounding

Module: POS | Severity: Medium

Reproduction: Use the cited current fixture workflow or inspect the relevant original evidence log; historical pre-fix reproduction was not reintroduced.

Expected: BRD requirement and invariant described by the finding; no duplicate/incorrect financial or stock posting.

Actual / fix status: Fixed and regression-tested.

Retest: Existing regression scenarios passed in fresh Step 13 run; exact method records follow.

Evidence: docs/testing/defect-register.md; historical storage/logs/ evidence; fresh step13-junit.xml

## D12-02 - MySQL invoice rollback fault-injection coverage

Module: Invoices | Severity: Low

Reproduction: Use the cited current fixture workflow or inspect the relevant original evidence log; historical pre-fix reproduction was not reintroduced.

Expected: BRD requirement and invariant described by the finding; no duplicate/incorrect financial or stock posting.

Actual / fix status: Portable model-event injection passes.

Retest: Existing regression scenarios passed in fresh Step 13 run; exact method records follow.

Evidence: docs/testing/defect-register.md; historical storage/logs/ evidence; fresh step13-junit.xml

## G12-01 - Missing inventory issue/receiving/adjustment/count operations

Module: Inventory | Severity: High

Reproduction: Use the cited current fixture workflow or inspect the relevant original evidence log; historical pre-fix reproduction was not reintroduced.

Expected: BRD requirement and invariant described by the finding; no duplicate/incorrect financial or stock posting.

Actual / fix status: Implemented and tested; full unused-issue return supported.

Retest: Existing regression scenarios passed in fresh Step 13 run; exact method records follow.

Evidence: docs/testing/defect-register.md; historical storage/logs/ evidence; fresh step13-junit.xml

## G12-02 - Missing suppliers/purchases/GRNs/settlement

Module: Suppliers/Purchases | Severity: High

Reproduction: Use the cited current fixture workflow or inspect the relevant original evidence log; historical pre-fix reproduction was not reintroduced.

Expected: BRD requirement and invariant described by the finding; no duplicate/incorrect financial or stock posting.

Actual / fix status: Implemented and tested, including opening settlement and return credits.

Retest: Existing regression scenarios passed in fresh Step 13 run; exact method records follow.

Evidence: docs/testing/defect-register.md; historical storage/logs/ evidence; fresh step13-junit.xml

## D12-03 - Approved quotation did not transfer job selling totals

Module: Quotations/Jobs | Severity: High

Reproduction: Use the cited current fixture workflow or inspect the relevant original evidence log; historical pre-fix reproduction was not reintroduced.

Expected: BRD requirement and invariant described by the finding; no duplicate/incorrect financial or stock posting.

Actual / fix status: Fixed; receipt-to-paid-invoice test uses real service calls, without manually setting job prices.

Retest: Existing regression scenarios passed in fresh Step 13 run; exact method records follow.

Evidence: docs/testing/defect-register.md; historical storage/logs/ evidence; fresh step13-junit.xml

## D12-04 - Completion/delivery statuses could leave timestamps empty

Module: Jobs | Severity: Medium

Reproduction: Use the cited current fixture workflow or inspect the relevant original evidence log; historical pre-fix reproduction was not reintroduced.

Expected: BRD requirement and invariant described by the finding; no duplicate/incorrect financial or stock posting.

Actual / fix status: Fixed; timestamp assertions pass.

Retest: Existing regression scenarios passed in fresh Step 13 run; exact method records follow.

Evidence: docs/testing/defect-register.md; historical storage/logs/ evidence; fresh step13-junit.xml

## D12-05 - Older returned stock could replace newer purchase cost

Module: Inventory | Severity: Medium

Reproduction: Use the cited current fixture workflow or inspect the relevant original evidence log; historical pre-fix reproduction was not reintroduced.

Expected: BRD requirement and invariant described by the finding; no duplicate/incorrect financial or stock posting.

Actual / fix status: Fixed with owner-approved latest-purchase source selection.

Retest: Existing regression scenarios passed in fresh Step 13 run; exact method records follow.

Evidence: docs/testing/defect-register.md; historical storage/logs/ evidence; fresh step13-junit.xml

## D12-06 - Accounting payment reversal reduced reported sales

Module: Reports | Severity: Medium

Reproduction: Use the cited current fixture workflow or inspect the relevant original evidence log; historical pre-fix reproduction was not reintroduced.

Expected: BRD requirement and invariant described by the finding; no duplicate/incorrect financial or stock posting.

Actual / fix status: Fixed; collections change while issued sale remains intact.

Retest: Existing regression scenarios passed in fresh Step 13 run; exact method records follow.

Evidence: docs/testing/defect-register.md; historical storage/logs/ evidence; fresh step13-junit.xml

## G12-04 - Intake PDF gaps, fonts, statement title/paper propagation

Module: Printing | Severity: Medium

Reproduction: Use the cited current fixture workflow or inspect the relevant original evidence log; historical pre-fix reproduction was not reintroduced.

Expected: BRD requirement and invariant described by the finding; no duplicate/incorrect financial or stock posting.

Actual / fix status: Implemented and tested; short fixture PDFs visually reviewed. Physical acceptance pending.

Retest: Existing regression scenarios passed in fresh Step 13 run; exact method records follow.

Evidence: docs/testing/defect-register.md; historical storage/logs/ evidence; fresh step13-junit.xml

## G12-03 - Operational backup destination/cloud account unset

Module: Backups | Severity: High

Reproduction: Use the cited current fixture workflow or inspect the relevant original evidence log; historical pre-fix reproduction was not reintroduced.

Expected: BRD requirement and invariant described by the finding; no duplicate/incorrect financial or stock posting.

Actual / fix status: Awaiting approved private folder; Drive remains blank by request. No host scheduler activated.

Retest: Not Run for external/unsupported boundary; not accepted by automation.

Evidence: docs/testing/defect-register.md; historical storage/logs/ evidence; fresh step13-junit.xml

## G12-05 - Purchase-wide cost allocation / ambiguous financial treatment

Module: Finance | Severity: Medium

Reproduction: Use the cited current fixture workflow or inspect the relevant original evidence log; historical pre-fix reproduction was not reintroduced.

Expected: BRD requirement and invariant described by the finding; no duplicate/incorrect financial or stock posting.

Actual / fix status: Latest purchase method approved; freight/discount allocation still awaits answer. Affected profit stays incomplete.

Retest: Not Run for external/unsupported boundary; not accepted by automation.

Evidence: docs/testing/defect-register.md; historical storage/logs/ evidence; fresh step13-junit.xml

## G12-06 - Reception/technician/repair-POS/GRN concurrency

Module: Shared concurrency | Severity: Medium

Reproduction: Use the cited current fixture workflow or inspect the relevant original evidence log; historical pre-fix reproduction was not reintroduced.

Expected: BRD requirement and invariant described by the finding; no duplicate/incorrect financial or stock posting.

Actual / fix status: Two-worker native cases pass. Realistic-volume capacity remains unmeasured.

Retest: Existing regression scenarios passed in fresh Step 13 run; exact method records follow.

Evidence: docs/testing/defect-register.md; historical storage/logs/ evidence; fresh step13-junit.xml

### G12-07 - Physical printing/scanning and business sign-off

Module: Hardware/UAT | Severity: Medium

Reproduction: Use the cited current fixture workflow or inspect the relevant original evidence log; historical pre-fix reproduction was not reintroduced.

Expected: BRD requirement and invariant described by the finding; no duplicate/incorrect financial or stock posting.

Actual / fix status: External UAT pending; no signatures fabricated.

Retest: Not Run for external/unsupported boundary; not accepted by automation.

Evidence: docs/testing/defect-register.md; historical storage/logs/ evidence; fresh step13-junit.xml

### L12-01 - Partial return of one repair issue / repeated-part historical purchase return

Module: Inventory/Purchases | Severity: Low

Reproduction: Use the cited current fixture workflow or inspect the relevant original evidence log; historical pre-fix reproduction was not reintroduced.

Expected: BRD requirement and invariant described by the finding; no duplicate/incorrect financial or stock posting.

Actual / fix status: Full unused repair issue can be reversed; ambiguous repeated-part purchase return is refused pending source reconciliation.

Retest: Not Run for external/unsupported boundary; not accepted by automation.

Evidence: docs/testing/defect-register.md; historical storage/logs/ evidence; fresh step13-junit.xml

### D13-01 - Unresolved English translation keys on quotation page

Module: Quotations/UI | Severity: Low

Reproduction: Open an issued quotation detail page in English; inspect customer contact and labour total labels.

Expected: Readable localized UI field labels.

Actual / fix status: Observed in actual screenshot; not fixed in this documentation-only task.

Retest: Not Run for external/unsupported boundary; not accepted by automation.

Evidence: documentation/screenshots/quotation-approval.png

# 19. Retesting

This fresh run re-executed the existing regression suite after the earlier Step 12 fixes; 218 cases passed. Historical failures and prior retest logs remain under storage/logs and docs/testing. They were not overwritten or reclassified as newly executed failures. New D13-01 was observed during capture; no fix/retest claim is made. Document layout corrections were regenerated and visually checked separately from application testing.

# 20. Known limitations

- Physical A5/A6/A4/thermal printing: Not Run. Actual shop devices unavailable; inspect paper/scale/signatures and both scripts.
- Physical barcode scanner: Not Run. Browser search/scan field tested; hardware keyboard-wedge scan not exercised.
- Full end-to-end browser business workflow: Not Run. Screenshots verify page rendering; backend scenarios verify posting. No claim of a full browser transaction/UAT test.

- Wall-clock session expiration: Not Run. Configured 120 minutes inspected; waiting through real expiry was not performed.
- Host unattended 02:00 scheduler execution: Not Run. No host scheduler activated by this task.
- Realistic-volume capacity: Not Run. Local fixture observations are not a load/capacity test.
- All hostile uploads and every XSS sink: Not Run. Targeted probes only; no exhaustive penetration test.
- Printer/scanner business-user sign-off: Not Run. Physical test and business tester/date/signature required.
- Live operational backup destination: Blocked. Approved private operational folder has not been supplied.
- Real Google Drive upload: Blocked. Folder/account/OAuth intentionally blank at owner request.
- Purchase-wide freight/discount policy acceptance: Blocked. Owner chose latest purchase cost; allocation of purchase-wide adjustments remains undecided.
- Final business-owner UAT approval: Blocked. No owner signature/acceptance supplied; automation cannot provide it.

Not Implemented UI boundaries: public self-registration/reset/general account settings, multi-invoice payment allocation form and partial return of one repair issue. Historical repeated-part purchase returns require reconciliation. Reported profit is an incomplete recorded-cost estimate where policy/cost data is missing. Source-presence does not imply live configuration or acceptance.

# 21. Final technical assessment

The executed isolated automated and native invariant checks passed for the current implementation. The operational metadata/count comparison matched and the screenshots came from separate temporary fixtures. Application acceptance remains conditional on the external checks and decisions above. No 100% BRD compliance, exhaustive security coverage, owner approval, deployment or live recovery is asserted.

# 22. Per-case execution records

Each data-set execution is a distinct JUnit case. Exact test method/source and raw evidence identify the executable fixture and assertion specification. These records are generated from that source and JUnit rather than invented outcomes.

### T13-001 - Credential key variants are redacted in both audit snapshots (#0)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Credential key variants are redacted in both audit snapshots (#0). Assert/check operations: assertSame, assertStringNotContainsString.

Actual: Passed; 4 assertions; 0.037705 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:24; Tests\Unit\AuditSafetyTest::test_credential_key_variants_are_redacted_in_both_audit_snapshots with data set #0; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-002 - Credential key variants are redacted in both audit snapshots (#1)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Credential key variants are redacted in both audit snapshots (#1). Assert/check operations: assertSame, assertStringNotContainsString.

Actual: Passed; 4 assertions; 0.002026 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:24; Tests\Unit\AuditSafetyTest::test_credential_key_variants_are_redacted_in_both_audit_snapshots with data set #1; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-003 - Credential key variants are redacted in both audit snapshots (#2)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Credential key variants are redacted in both audit snapshots (#2). Assert/check operations: assertSame, assertStringNotContainsString.

Actual: Passed; 4 assertions; 0.002547 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:24; Tests\Unit\AuditSafetyTest::test_credential_key_variants_are_redacted_in_both_audit_snapshots with data set #2; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-004 - Credential key variants are redacted in both audit snapshots (#3)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Credential key variants are redacted in both audit snapshots (#3). Assert/check operations: assertSame, assertStringNotContainsString.

Actual: Passed; 4 assertions; 0.002660 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:24; Tests\Unit\AuditSafetyTest::test_credential_key_variants_are_redacted_in_both_audit_snapshots with data set #3; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-005 - Credential key variants are redacted in both audit snapshots (#4)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Credential key variants are redacted in both audit snapshots (#4). Assert/check operations: assertSame, assertStringNotContainsString.

Actual: Passed; 4 assertions; 0.002251 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:24; Tests\Unit\AuditSafetyTest::test_credential_key_variants_are_redacted_in_both_audit_snapshots with data set #4; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-006 - Credential key variants are redacted in both audit snapshots (#5)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Credential key variants are redacted in both audit snapshots (#5). Assert/check operations: assertSame, assertStringNotContainsString.

Actual: Passed; 4 assertions; 0.001824 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:24; Tests\Unit\AuditSafetyTest::test_credential_key_variants_are_redacted_in_both_audit_snapshots with data set #5; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-007 - Credential key variants are redacted in both audit snapshots (#6)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Credential key variants are redacted in both audit snapshots (#6). Assert/check operations: assertSame, assertStringNotContainsString.

Actual: Passed; 4 assertions; 0.001649 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:24; Tests\Unit\AuditSafetyTest::test_credential_key_variants_are_redacted_in_both_audit_snapshots with data set #6; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-008 - Credential key variants are redacted in both audit snapshots (#7)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Credential key variants are redacted in both audit snapshots (#7). Assert/check operations: assertSame, assertStringNotContainsString.

Actual: Passed; 4 assertions; 0.002237 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:24; Tests\Unit\AuditSafetyTest::test_credential_key_variants_are_redacted_in_both_audit_snapshots with data set #7; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-009 - Credential key variants are redacted in both audit snapshots (#8)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Credential key variants are redacted in both audit snapshots (#8). Assert/check operations: assertSame, assertStringNotContainsString.

Actual: Passed; 4 assertions; 0.002270 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:24; Tests\Unit\AuditSafetyTest::test_credential_key_variants_are_redacted_in_both_audit_snapshots with data set #8; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-010 - Credential key variants are redacted in both audit snapshots (#9)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Credential key variants are redacted in both audit snapshots (#9). Assert/check operations: assertSame, assertStringNotContainsString.

Actual: Passed; 4 assertions; 0.001547 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:24; Tests\Unit\AuditSafetyTest::test_credential_key_variants_are_redacted_in_both_audit_snapshots with data set #9; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-011 - Credential key variants are redacted in both audit snapshots (#10)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Credential key variants are redacted in both audit snapshots (#10). Assert/check operations: assertSame, assertStringNotContainsString.

Actual: Passed; 4 assertions; 0.001496 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:24; Tests\Unit\AuditSafetyTest::test_credential_key_variants_are_redacted_in_both_audit_snapshots with data set #10; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-012 - Credential key variants are redacted in both audit snapshots (#11)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Credential key variants are redacted in both audit snapshots (#11). Assert/check operations: assertSame, assertStringNotContainsString.

Actual: Passed; 4 assertions; 0.001485 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:24; Tests\Unit\AuditSafetyTest::test_credential_key_variants_are_redacted_in_both_audit_snapshots with data set #11; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-013 - Credential key variants are redacted in both audit snapshots (#12)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Credential key variants are redacted in both audit snapshots (#12). Assert/check operations: assertSame, assertStringNotContainsString.

Actual: Passed; 4 assertions; 0.001574 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:24; Tests\Unit\AuditSafetyTest::test_credential_key_variants_are_redacted_in_both_audit_snapshots with data set #12; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-014 - Credential key variants are redacted in both audit snapshots (#13)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Credential key variants are redacted in both audit snapshots (#13). Assert/check operations: assertSame, assertStringNotContainsString.

Actual: Passed; 4 assertions; 0.001703 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:24; Tests\Unit\AuditSafetyTest::test_credential_key_variants_are_redacted_in_both_audit_snapshots with data set #13; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-015 - Preencoded json is redacted once and null remains null

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Preencoded json is redacted once and null remains null. Assert/check operations: assertSame, assertIsArray, assertNull.

Actual: Passed; 4 assertions; 0.001473 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:45; Tests\Unit\AuditSafetyTest::test_preencoded_json_is_redacted_once_and_null_remains_null; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-016 - Hydrated legacy audit values are redacted when read

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Hydrated legacy audit values are redacted when read. Assert/check operations: assertSame.

Actual: Passed; 2 assertions; 0.001392 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:59; Tests\Unit\AuditSafetyTest::test_hydrated_legacy_audit_values_are_redacted_when_read; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-017 - Unrelated business fields and scalar settings are preserved

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Unrelated business fields and scalar settings are preserved. Assert/check operations: assertSame, assertSafeForStorage.

Actual: Passed; 6 assertions; 0.002523 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:68; Tests\Unit\AuditSafetyTest::test_unrelated_business_fields_and_scalar_settings_are_preserved; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-018 - Settings reject credential names on assignment (#0)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and expectException; no operational business values.

Execution steps: Set up/use fixture helpers: expectException.

Expected: Settings reject credential names on assignment (#0). Assert/check operations: expectException.

Actual: Passed; 1 assertions; 0.001454 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:86; Tests\Unit\AuditSafetyTest::test_settings_reject_credential_names_on_assignment with data set #0; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-019 - Settings reject credential names on assignment (#1)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and expectException; no operational business values.

Execution steps: Set up/use fixture helpers: expectException.

Expected: Settings reject credential names on assignment (#1). Assert/check operations: expectException.

Actual: Passed; 1 assertions; 0.001380 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:86; Tests\Unit\AuditSafetyTest::test_settings_reject_credential_names_on_assignment with data set #1; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-020 - Settings reject credential names on assignment (#2)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and expectException; no operational business values.

Execution steps: Set up/use fixture helpers: expectException.

Expected: Settings reject credential names on assignment (#2). Assert/check operations: expectException.

Actual: Passed; 1 assertions; 0.001408 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:86; Tests\Unit\AuditSafetyTest::test_settings_reject_credential_names_on_assignment with data set #2; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-021 - Settings reject credential names on assignment (#3)

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and expectException; no operational business values.

Execution steps: Set up/use fixture helpers: expectException.

Expected: Settings reject credential names on assignment (#3). Assert/check operations: expectException.

Actual: Passed; 1 assertions; 0.001415 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:86; Tests\Unit\AuditSafetyTest::test_settings_reject_credential_names_on_assignment with data set #3; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-022 - Settings reject nested sensitive fields in json strings

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and expectException; no operational business values.

Execution steps: Set up/use fixture helpers: expectException.

Expected: Settings reject nested sensitive fields in json strings. Assert/check operations: expectException.

Actual: Passed; 1 assertions; 0.002102 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:98; Tests\Unit\AuditSafetyTest::test_settings_reject_nested_sensitive_fields_in_json_strings; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-023 - Saving rechecks unsafe raw attributes without a database

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and expectException; no operational business values.

Execution steps: Set up/use fixture helpers: expectException.

Expected: Saving rechecks unsafe raw attributes without a database. Assert/check operations: expectException.

Actual: Passed; 1 assertions; 0.008977 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Unit/AuditSafetyTest.php:105; Tests\Unit\AuditSafetyTest::test_saving_rechecks_unsafe_raw_attributes_without_a_database; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-024 - Login screen renders a csrf protected form

Module: Authentication | BRD section(s): 14, 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute requests in method order: GET login.

Expected: Login screen renders a csrf protected form. Assert/check operations: assertOk, assertSee, assertDontSee.

Actual: Passed; 5 assertions; 4.414224 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Auth/AuthenticationTest.php:13; Tests\Feature\Auth\AuthenticationTest::test_login_screen_renders_a_csrf_protected_form; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-025 - Users can log in and the session id is regenerated

Module: Authentication | BRD section(s): 14, 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and withSession; no operational business values.

Execution steps: Set up/use fixture helpers: withSession. Execute requests in method order: GET login; POST login.store.

Expected: Users can log in and the session id is regenerated. Assert/check operations: assertSessionHasNoErrors, assertSessionHas, assertRedirect, assertAuthenticatedAs, assertNotSame.

Actual: Passed; 8 assertions; 0.605395 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Auth/AuthenticationTest.php:23; Tests\Feature\Auth\AuthenticationTest::test_users_can_log_in_and_the_session_id_is_regenerated; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-026 - Users cannot log in with an invalid password

Module: Authentication | BRD section(s): 14, 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and from; no operational business values.

Execution steps: Set up/use fixture helpers: from. Execute requests in method order: POST login.store.

Expected: Users cannot log in with an invalid password. Assert/check operations: assertRedirect, assertSessionHasErrors, assertGuest.

Actual: Passed; 5 assertions; 0.272951 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Auth/AuthenticationTest.php:43; Tests\Feature\Auth\AuthenticationTest::test_users_cannot_log_in_with_an_invalid_password; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-027 - Login requires a username and password

Module: Authentication | BRD section(s): 14, 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute requests in method order: POST login.store.

Expected: Login requires a username and password. Assert/check operations: assertSessionHasErrors, assertGuest.

Actual: Passed; 4 assertions; 0.041446 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Auth/AuthenticationTest.php:55; Tests\Feature\Auth\AuthenticationTest::test_login_requires_a_username_and_password; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-028 - Login rejects non string credentials

Module: Authentication | BRD section(s): 14, 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute requests in method order: POST login.store.

Expected: Login rejects non string credentials. Assert/check operations: assertSessionHasErrors, assertGuest.

Actual: Passed; 4 assertions; 0.048213 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Auth/AuthenticationTest.php:63; Tests\Feature\Auth\AuthenticationTest::test_login_rejects_non_string_credentials; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-029 - Login is throttled after five attempts per username and ip

Module: Authentication | BRD section(s): 14, 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and travel; no operational business values.

Execution steps: Set up/use fixture helpers: travel. Execute requests in method order: POST login.store; POST login.store; POST login.store.

Expected: Login is throttled after five attempts per username and ip. Assert/check operations: assertSessionHasErrors, assertTooManyRequests, assertGuest, assertRedirect, assertAuthenticatedAs.

Actual: Passed; 17 assertions; 1.480912 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Auth/AuthenticationTest.php:73; Tests\Feature\Auth\AuthenticationTest::test_login_is_throttled_after_five_attempts_per_username_and_ip; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-030 - Logout invalidates the session and regenerates the csrf token

Module: Authentication | BRD section(s): 14, 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and withSession; no operational business values.

Execution steps: Set up/use fixture helpers: withSession. Execute requests in method order: GET login; POST login.store; POST logout.

Expected: Logout invalidates the session and regenerates the csrf token. Assert/check operations: assertRedirect, assertSessionMissing, assertGuest, assertNotSame.

Actual: Passed; 6 assertions; 0.272739 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Auth/AuthenticationTest.php:101; Tests\Feature\Auth\AuthenticationTest::test_logout_invalidates_the_session_and_regenerates_the_csrf_token; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-031 - Login rejects requests without a csrf token when verification is active

Module: Authentication | BRD section(s): 14, 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and withSession; no operational business values.

Execution steps: Set up/use fixture helpers: withSession. Execute requests in method order: POST login.store; POST login.store.

Expected: Login rejects requests without a csrf token when verification is active. Assert/check operations: assertStatus, assertGuest, assertRedirect, assertAuthenticatedAs.

Actual: Passed; 7 assertions; 0.280693 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Auth/AuthenticationTest.php:124; Tests\Feature\Auth\AuthenticationTest::test_login_rejects_requests_without_a_csrf_token_when_verification_is_active; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-032 - Unused authentication features are not exposed

Module: Authentication | BRD section(s): 14, 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Unused authentication features are not exposed. Assert/check operations: assertNotFound.

Actual: Passed; 7 assertions; 0.060339 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Auth/AuthenticationTest.php:148; Tests\Feature\Auth\AuthenticationTest::test_unused_authentication_features_are_not_exposed; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-033 - Public registration is not available

Module: Authentication | BRD section(s): 14, 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute requests in method order: GET /register; POST /register.

Expected: Public registration is not available. Assert/check operations: assertNotFound, assertDatabaseCount.

Actual: Passed; 3 assertions; 0.036628 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Auth/AuthenticationTest.php:161; Tests\Feature\Auth\AuthenticationTest::test_public_registration_is_not_available; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-034 - Disabled accounts cannot log in

Module: Authentication | BRD section(s): 14, 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute requests in method order: POST login.store.

Expected: Disabled accounts cannot log in. Assert/check operations: assertSessionHasErrors, assertGuest.

Actual: Passed; 3 assertions; 0.039629 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Auth/AuthenticationTest.php:175; Tests\Feature\Auth\AuthenticationTest::test_disabled_accounts_cannot_log_in; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-035 - Backup destination is disabled until private mysql tools are configured

Module: Backups | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed; no operational business values.

Execution steps: Set up/use fixture helpers: seed. Execute requests in method order: GET backups.index; POST backups.create.

Expected: Backup destination is disabled until private mysql tools are configured. Assert/check operations: assertOk, assertSee, assertRedirect, assertSessionHasErrors, assertDatabaseCount.

Actual: Passed; 6 assertions; 0.930983 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/BackupManagementTest.php:14; Tests\Feature\BackupManagementTest::test_backup_destination_is_disabled_until_private_mysql_tools_are_configured; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-036 - Database restore is not available over http

Module: Backups | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed; no operational business values.

Execution steps: Set up/use fixture helpers: seed. Execute requests in method order: POST /backups/1/restore.

Expected: Database restore is not available over http. Assert/check operations: assertNotFound.

Actual: Passed; 1 assertions; 1.450715 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/BackupManagementTest.php:27; Tests\Feature\BackupManagementTest::test_database_restore_is_not_available_over_http; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-037 - Private bootstrap creates a linked administrator with a hashed password

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed, artisan; no operational business values.

Execution steps: Set up/use fixture helpers: seed, artisan.

Expected: Private bootstrap creates a linked administrator with a hashed password. Assert/check operations: assertSuccessful, assertSame, assertTrue, assertNotNull, assertNotSame, assertArrayNotHasKey.

Actual: Passed; 15 assertions; 1.816809 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CreateUserCommandTest.php:15; Tests\Feature\CreateUserCommandTest::test_private_bootstrap_creates_a_linked_administrator_with_a_hashed_password; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-038 - Duplicate usernames are rejected before creating an account

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed, artisan; no operational business values.

Execution steps: Set up/use fixture helpers: seed, artisan.

Expected: Duplicate usernames are rejected before creating an account. Assert/check operations: assertFailed, assertDatabaseCount.

Actual: Passed; 7 assertions; 0.777435 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CreateUserCommandTest.php:41; Tests\Feature\CreateUserCommandTest::test_duplicate_usernames_are_rejected_before_creating_an_account; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-039 - Unconfirmed passwords are rejected

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed, artisan; no operational business values.

Execution steps: Set up/use fixture helpers: seed, artisan.

Expected: Unconfirmed passwords are rejected. Assert/check operations: assertFailed, assertDatabaseCount.

Actual: Passed; 7 assertions; 0.718739 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CreateUserCommandTest.php:57; Tests\Feature\CreateUserCommandTest::test_unconfirmed_passwords_are_rejected; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-040 - Passwords shorter than twelve characters are rejected

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed, artisan; no operational business values.

Execution steps: Set up/use fixture helpers: seed, artisan.

Expected: Passwords shorter than twelve characters are rejected. Assert/check operations: assertFailed, assertDatabaseCount.

Actual: Passed; 7 assertions; 0.768654 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CreateUserCommandTest.php:71; Tests\Feature\CreateUserCommandTest::test_passwords_shorter_than_twelve_characters_are_rejected; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-041 - Invalid identity fields are rejected

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed, artisan; no operational business values.

Execution steps: Set up/use fixture helpers: seed, artisan.

Expected: Invalid identity fields are rejected. Assert/check operations: assertFailed, assertDatabaseCount.

Actual: Passed; 7 assertions; 0.784348 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CreateUserCommandTest.php:85; Tests\Feature\CreateUserCommandTest::test_invalid_identity_fields_are_rejected; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-042 - The default seeder does not create demo accounts

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed; no operational business values.

Execution steps: Set up/use fixture helpers: seed.

Expected: The default seeder does not create demo accounts. Assert/check operations: assertDatabaseCount.

Actual: Passed; 1 assertions; 0.041244 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CreateUserCommandTest.php:99; Tests\Feature\CreateUserCommandTest::test_the_default_seeder_does_not_create_demo_accounts; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-043 - Receptionist can register individual and company customers

Module: Customers | BRD section(s): 4 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, validIndividual, validCompany; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, validIndividual, validCompany. Execute requests in method order: POST customers.store; POST customers.store.

Expected: Receptionist can register individual and company customers. Assert/check operations: assertRedirect, assertSame, assertDatabaseCount, assertNotSame.

Actual: Passed; 11 assertions; 1.014908 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CustomerManagementTest.php:22; Tests\Feature\CustomerManagementTest::test_receptionist_can_register_individual_and_company_customers; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-044 - Authorized financial user records opening balance once

Module: Customers | BRD section(s): 4 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, validIndividual; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, validIndividual. Execute requests in method order: POST customers.store.

Expected: Authorized financial user records opening balance once. Assert/check operations: assertRedirect, assertSame, assertDatabaseHas.

Actual: Passed; 3 assertions; 1.006949 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CustomerManagementTest.php:46; Tests\Feature\CustomerManagementTest::test_authorized_financial_user_records_opening_balance_once; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-045 - Zero opening balance does not create a ledger entry

Module: Customers | BRD section(s): 4 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, validIndividual; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, validIndividual. Execute requests in method order: POST customers.store.

Expected: Zero opening balance does not create a ledger entry. Assert/check operations: assertRedirect, assertDatabaseCount.

Actual: Passed; 2 assertions; 0.837337 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CustomerManagementTest.php:64; Tests\Feature\CustomerManagementTest::test_zero_opening_balance_does_not_create_a_ledger_entry; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-046 - Customer forms show only fields allowed by the users permissions

Module: Customers | BRD section(s): 4 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole. Execute requests in method order: GET customers.create; GET customers.create.

Expected: Customer forms show only fields allowed by the users permissions. Assert/check operations: assertOk, assertDontSee, assertSee.

Actual: Passed; 6 assertions; 0.951444 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CustomerManagementTest.php:74; Tests\Feature\CustomerManagementTest::test_customer_forms_show_only_fields_allowed_by_the_users_permissions; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-047 - Receptionist cannot submit financial balance or inactive status

Module: Customers | BRD section(s): 4 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, validIndividual; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, validIndividual. Execute requests in method order: POST customers.store.

Expected: Receptionist cannot submit financial balance or inactive status. Assert/check operations: assertSessionHasErrors, assertDatabaseCount.

Actual: Passed; 4 assertions; 0.857855 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CustomerManagementTest.php:89; Tests\Feature\CustomerManagementTest::test_receptionist_cannot_submit_financial_balance_or_inactive_status; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-048 - Customer validation requires type name address and valid contact data

Module: Customers | BRD section(s): 4 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, validIndividual; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, validIndividual. Execute requests in method order: POST customers.store; POST customers.store; POST customers.store.

Expected: Customer validation requires type name address and valid contact data. Assert/check operations: assertRedirect, assertSessionHasErrors, assertDatabaseCount.

Actual: Passed; 13 assertions; 0.854453 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CustomerManagementTest.php:101; Tests\Feature\CustomerManagementTest::test_customer_validation_requires_type_name_address_and_valid_contact_data; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-049 - Company type requires company name but email and contact person are optional

Module: Customers | BRD section(s): 4 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, validCompany; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, validCompany. Execute requests in method order: POST customers.store; POST customers.store.

Expected: Company type requires company name but email and contact person are optional. Assert/check operations: assertSessionHasErrors, assertRedirect, assertNull.

Actual: Passed; 5 assertions; 0.826930 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CustomerManagementTest.php:127; Tests\Feature\CustomerManagementTest::test_company_type_requires_company_name_but_email_and_contact_person_are_optional; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-050 - Customer search matches customer code name and mobile and excludes inactive

Module: Customers | BRD section(s): 4 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, registerCustomer, validIndividual; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, registerCustomer, validIndividual.

Expected: Customer search matches customer code name and mobile and excludes inactive. Assert/check operations: assertOk, assertJsonPath, assertJsonCount.

Actual: Passed; 10 assertions; 0.859290 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CustomerManagementTest.php:145; Tests\Feature\CustomerManagementTest::test_customer_search_matches_customer_code_name_and_mobile_and_excludes_inactive; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-051 - Customer list searches company name filters and paginates in database

Module: Customers | BRD section(s): 4 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, registerCustomer, validCompany, validIndividual; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, registerCustomer, validCompany, validIndividual. Execute requests in method order: GET customers.index; GET customers.index.

Expected: Customer list searches company name filters and paginates in database. Assert/check operations: assertOk, assertSee.

Actual: Passed; 49 assertions; 1.267546 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CustomerManagementTest.php:166; Tests\Feature\CustomerManagementTest::test_customer_list_searches_company_name_filters_and_paginates_in_database; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-052 - Customer can be updated without changing id or opening balance

Module: Customers | BRD section(s): 4 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, registerCustomer, validIndividual; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, registerCustomer, validIndividual. Execute requests in method order: PUT customers.update.

Expected: Customer can be updated without changing id or opening balance. Assert/check operations: assertRedirect, assertSame, assertDatabaseCount, assertDatabaseHas.

Actual: Passed; 10 assertions; 0.805698 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CustomerManagementTest.php:188; Tests\Feature\CustomerManagementTest::test_customer_can_be_updated_without_changing_id_or_opening_balance; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-053 - Customer model prevents id code and opening balance changes

Module: Customers | BRD section(s): 4 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, registerCustomer, validIndividual, expectException; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, registerCustomer, validIndividual, expectException.

Expected: Customer model prevents id code and opening balance changes. Assert/check operations: expectException.

Actual: Passed; 3 assertions; 0.898530 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CustomerManagementTest.php:218; Tests\Feature\CustomerManagementTest::test_customer_model_prevents_id_code_and_opening_balance_changes; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-054 - Deactivation requires permission preserves history and activation uses update permission

Module: Customers | BRD section(s): 4 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, registerCustomer, validIndividual; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, registerCustomer, validIndividual. Execute requests in method order: POST customers.deactivate; POST customers.deactivate; POST customers.activate.

Expected: Deactivation requires permission preserves history and activation uses update permission. Assert/check operations: assertForbidden, assertTrue, assertRedirect, assertFalse, assertDatabaseHas.

Actual: Passed; 11 assertions; 0.873520 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CustomerManagementTest.php:227; Tests\Feature\CustomerManagementTest::test_deactivation_requires_permission_preserves_history_and_activation_uses_update_permission; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-055 - Customer statement and financial fields require statement permission

Module: Customers | BRD section(s): 4 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, registerCustomer, validIndividual; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, registerCustomer, validIndividual. Execute requests in method order: GET customers.show; GET customers.statement; GET customers.show; GET customers.statement.

Expected: Customer statement and financial fields require statement permission. Assert/check operations: assertOk, assertDontSee, assertForbidden, assertSee.

Actual: Passed; 9 assertions; 1.096365 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CustomerManagementTest.php:249; Tests\Feature\CustomerManagementTest::test_customer_statement_and_financial_fields_require_statement_permission; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-056 - Unauthorized registration and updates are blocked

Module: Customers | BRD section(s): 4 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and registerCustomer, userWithRole, validIndividual; no operational business values.

Execution steps: Set up/use fixture helpers: registerCustomer, userWithRole, validIndividual. Execute requests in method order: GET customers.index; POST customers.store; PUT customers.update.

Expected: Unauthorized registration and updates are blocked. Assert/check operations: assertForbidden, assertDatabaseCount.

Actual: Passed; 6 assertions; 0.859737 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/CustomerManagementTest.php:272; Tests\Feature\CustomerManagementTest::test_unauthorized_registration_and_updates_are_blocked; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-057 - Home redirects to the dashboard

Module: Dashboard | BRD section(s): 18 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute requests in method order: GET /.

Expected: Home redirects to the dashboard. Assert/check operations: assertRedirect.

Actual: Passed; 2 assertions; 0.036734 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/DashboardTest.php:14; Tests\Feature\DashboardTest::test_home_redirects_to_the_dashboard; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-058 - Guests are redirected to login

Module: Dashboard | BRD section(s): 18 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute requests in method order: GET dashboard.

Expected: Guests are redirected to login. Assert/check operations: assertRedirect.

Actual: Passed; 2 assertions; 0.049557 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/DashboardTest.php:19; Tests\Feature\DashboardTest::test_guests_are_redirected_to_login; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-059 - Authenticated users can view the dashboard without email verification

Module: Dashboard | BRD section(s): 18 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed; no operational business values.

Execution steps: Set up/use fixture helpers: seed. Execute requests in method order: GET dashboard.

Expected: Authenticated users can view the dashboard without email verification. Assert/check operations: assertOk, assertSee.

Actual: Passed; 8 assertions; 0.942697 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/DashboardTest.php:24; Tests\Feature\DashboardTest::test_authenticated_users_can_view_the_dashboard_without_email_verification; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-060 - Unimplemented business modules do not have working crud endpoints

Module: Dashboard | BRD section(s): 18 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute requests in method order: GET /; POST /.

Expected: Unimplemented business modules do not have working crud endpoints. Assert/check operations: assertNotFound.

Actual: Passed; 6 assertions; 0.051779 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/DashboardTest.php:41; Tests\Feature\DashboardTest::test_unimplemented_business_modules_do_not_have_working_crud_endpoints; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-061 - Health endpoint is available

Module: Dashboard | BRD section(s): 18 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute requests in method order: GET /up.

Expected: Health endpoint is available. Assert/check operations: assertOk.

Actual: Passed; 1 assertions; 0.032447 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/DashboardTest.php:55; Tests\Feature\DashboardTest::test_health_endpoint_is_available; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-062 - Reports require the reports view permission

Module: Dashboard | BRD section(s): 18 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed; no operational business values.

Execution steps: Set up/use fixture helpers: seed. Execute requests in method order: GET /reports; GET reports.show.

Expected: Reports require the reports view permission. Assert/check operations: assertOk, assertForbidden.

Actual: Passed; 2 assertions; 0.978670 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/DashboardTest.php:60; Tests\Feature\DashboardTest::test_reports_require_the_reports_view_permission; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-063 - Job numbers cannot be duplicated

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and job; no operational business values.

Execution steps: Set up/use fixture helpers: job.

Expected: Job numbers cannot be duplicated. Assert/check operations: assertDatabaseRejects.

Actual: Passed; 1 assertions; 0.059276 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:22; Tests\Feature\Database\BusinessConstraintsTest::test_job_numbers_cannot_be_duplicated; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-064 - Invoice numbers cannot be duplicated

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and invoice; no operational business values.

Execution steps: Set up/use fixture helpers: invoice.

Expected: Invoice numbers cannot be duplicated. Assert/check operations: assertDatabaseRejects.

Actual: Passed; 1 assertions; 0.062415 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:29; Tests\Feature\Database\BusinessConstraintsTest::test_invoice_numbers_cannot_be_duplicated; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-065 - Quotation numbers cannot be duplicated

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and quotation; no operational business values.

Execution steps: Set up/use fixture helpers: quotation.

Expected: Quotation numbers cannot be duplicated. Assert/check operations: assertDatabaseRejects.

Actual: Passed; 1 assertions; 0.079773 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:36; Tests\Feature\Database\BusinessConstraintsTest::test_quotation_numbers_cannot_be_duplicated; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-066 - Customer codes and skus cannot be duplicated

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and customer, part; no operational business values.

Execution steps: Set up/use fixture helpers: customer, part.

Expected: Customer codes and skus cannot be duplicated. Assert/check operations: assertDatabaseRejects.

Actual: Passed; 2 assertions; 0.058641 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:43; Tests\Feature\Database\BusinessConstraintsTest::test_customer_codes_and_skus_cannot_be_duplicated; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-067 - Optional barcodes allow nulls but nonnull values are unique

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and part; no operational business values.

Execution steps: Set up/use fixture helpers: part.

Expected: Optional barcodes allow nulls but nonnull values are unique. Assert/check operations: assertDatabaseRejects.

Actual: Passed; 1 assertions; 0.054335 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:52; Tests\Feature\Database\BusinessConstraintsTest::test_optional_barcodes_allow_nulls_but_nonnull_values_are_unique; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-068 - Both composite foreign keys reject cross customer invoice job links

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and customer, invoice, job; no operational business values.

Execution steps: Set up/use fixture helpers: customer, invoice, job.

Expected: Both composite foreign keys reject cross customer invoice job links. Assert/check operations: assertDatabaseRejects.

Actual: Passed; 2 assertions; 0.040307 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:61; Tests\Feature\Database\BusinessConstraintsTest::test_both_composite_foreign_keys_reject_cross_customer_invoice_job_links; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-069 - One job cannot be reserved for billing by two invoices

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and customer, job, invoice; no operational business values.

Execution steps: Set up/use fixture helpers: customer, job, invoice.

Expected: One job cannot be reserved for billing by two invoices. Assert/check operations: assertDatabaseRejects.

Actual: Passed; 1 assertions; 0.063300 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:78; Tests\Feature\Database\BusinessConstraintsTest::test_one_job_cannot_be_reserved_for_billing_by_two_invoices; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-070 - Orphan foreign keys and parent deletion are rejected by the database

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and customer, job, invoice, code, employee; no operational business values.

Execution steps: Set up/use fixture helpers: customer, job, invoice, code, employee.

Expected: Orphan foreign keys and parent deletion are rejected by the database. Assert/check operations: assertDatabaseRejects.

Actual: Passed; 3 assertions; 0.043163 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:97; Tests\Feature\Database\BusinessConstraintsTest::test_orphan_foreign_keys_and_parent_deletion_are_rejected_by_the_database; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-071 - Customer type requires the corresponding name and nonblank contact fields

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and customer; no operational business values.

Execution steps: Set up/use fixture helpers: customer.

Expected: Customer type requires the corresponding name and nonblank contact fields. Assert/check operations: assertDatabaseRejects, assertSame.

Actual: Passed; 5 assertions; 0.034418 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:119; Tests\Feature\Database\BusinessConstraintsTest::test_customer_type_requires_the_corresponding_name_and_nonblank_contact_fields; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-072 - Nonpositive payments and invalid header totals are rejected

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and payment, invoice; no operational business values.

Execution steps: Set up/use fixture helpers: payment, invoice.

Expected: Nonpositive payments and invalid header totals are rejected. Assert/check operations: assertDatabaseRejects.

Actual: Passed; 3 assertions; 0.038042 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:135; Tests\Feature\Database\BusinessConstraintsTest::test_nonpositive_payments_and_invalid_header_totals_are_rejected; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-073 - Payment and allocation idempotency keys are unique

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and customer, payment, invoice, code; no operational business values.

Execution steps: Set up/use fixture helpers: customer, payment, invoice, code.

Expected: Payment and allocation idempotency keys are unique. Assert/check operations: assertDatabaseRejects.

Actual: Passed; 2 assertions; 0.062090 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:142; Tests\Feature\Database\BusinessConstraintsTest::test_payment_and_allocation_idempotency_keys_are_unique; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-074 - Payment allocations cannot cross customer boundaries or name multiple targets

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and customer, payment, invoice, code, job; no operational business values.

Execution steps: Set up/use fixture helpers: customer, payment, invoice, code, job.

Expected: Payment allocations cannot cross customer boundaries or name multiple targets. Assert/check operations: assertDatabaseRejects.

Actual: Passed; 3 assertions; 0.042191 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:166; Tests\Feature\Database\BusinessConstraintsTest::test_payment_allocations_cannot_cross_customer_boundaries_or_name_multiple_targets; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-075 - Stock operations reject duplicate keys duplicate issue sources and invalid signs

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and part, location, movement, job, code; no operational business values.

Execution steps: Set up/use fixture helpers: part, location, movement, job, code.

Expected: Stock operations reject duplicate keys duplicate issue sources and invalid signs. Assert/check operations: assertDatabaseRejects.

Actual: Passed; 4 assertions; 0.069592 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:205; Tests\Feature\Database\BusinessConstraintsTest::test_stock_operations_reject_duplicate_keys_duplicate_issue_sources_and_invalid_signs; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-076 - Job identifiers cannot be changed through the model

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and job, expectException; no operational business values.

Execution steps: Set up/use fixture helpers: job, expectException.

Expected: Job identifiers cannot be changed through the model. Assert/check operations: expectException.

Actual: Passed; 1 assertions; 0.067288 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:236; Tests\Feature\Database\BusinessConstraintsTest::test_job_identifiers_cannot_be_changed_through_the_model; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-077 - Status history cannot be updated through the model

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and statusHistory, expectException; no operational business values.

Execution steps: Set up/use fixture helpers: statusHistory, expectException.

Expected: Status history cannot be updated through the model. Assert/check operations: expectException.

Actual: Passed; 1 assertions; 0.043911 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:245; Tests\Feature\Database\BusinessConstraintsTest::test_status_history_cannot_be_updated_through_the_model; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-078 - Status history cannot be deleted through the model

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and statusHistory, expectException; no operational business values.

Execution steps: Set up/use fixture helpers: statusHistory, expectException.

Expected: Status history cannot be deleted through the model. Assert/check operations: expectException.

Actual: Passed; 1 assertions; 0.048773 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:254; Tests\Feature\Database\BusinessConstraintsTest::test_status_history_cannot_be_deleted_through_the_model; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-079 - Nullable customer ids cannot bypass payment owner consistency

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and customer, payment, code, employee; no operational business values.

Execution steps: Set up/use fixture helpers: customer, payment, code, employee.

Expected: Nullable customer ids cannot bypass payment owner consistency. Assert/check operations: assertDatabaseRejects.

Actual: Passed; 2 assertions; 0.061874 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:263; Tests\Feature\Database\BusinessConstraintsTest::test_nullable_customer_ids_cannot_bypass_payment_owner_consistency; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-080 - Allocation releases cannot use a reversal of another payment

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and customer, payment, invoice, code; no operational business values.

Execution steps: Set up/use fixture helpers: customer, payment, invoice, code.

Expected: Allocation releases cannot use a reversal of another payment. Assert/check operations: assertDatabaseRejects.

Actual: Passed; 1 assertions; 0.051159 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:300; Tests\Feature\Database\BusinessConstraintsTest::test_allocation_releases_cannot_use_a_reversal_of_another_payment; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-081 - Stock movements cannot use a source from another location

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and part, location, job, code, movement; no operational business values.

Execution steps: Set up/use fixture helpers: part, location, job, code, movement.

Expected: Stock movements cannot use a source from another location. Assert/check operations: assertDatabaseRejects.

Actual: Passed; 1 assertions; 0.054809 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:341; Tests\Feature\Database\BusinessConstraintsTest::test_stock_movements_cannot_use_a_source_from_another_location; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-082 - The job status enum matches all seventeen brd statuses

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: The job status enum matches all seventeen brd statuses. Assert/check operations: assertSame.

Actual: Passed; 1 assertions; 0.038150 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessConstraintsTest.php:361; Tests\Feature\Database\BusinessConstraintsTest::test_the_job_status_enum_matches_all_seventeen_brd_statuses; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-083 - Customers relate to their jobs receipts invoices payments and account entries

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and customer, job, code, invoice, payment; no operational business values.

Execution steps: Set up/use fixture helpers: customer, job, code, invoice, payment.

Expected: Customers relate to their jobs receipts invoices payments and account entries. Assert/check operations: assertTrue, assertSame.

Actual: Passed; 9 assertions; 0.065577 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessRelationshipsTest.php:28; Tests\Feature\Database\BusinessRelationshipsTest::test_customers_relate_to_their_jobs_receipts_invoices_payments_and_account_entries; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-084 - Repair jobs retain multiple status history inspection and parts usage records

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and employee, job, part, location, code; no operational business values.

Execution steps: Set up/use fixture helpers: employee, job, part, location, code.

Expected: Repair jobs retain multiple status history inspection and parts usage records. Assert/check operations: assertEqualsCanonicalizing, assertTrue, assertSame.

Actual: Passed; 4 assertions; 0.051966 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessRelationshipsTest.php:63; Tests\Feature\Database\BusinessRelationshipsTest::test_repair_jobs_retain_multiple_status_history_inspection_and_parts_usage_records; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-085 - One invoice retains multiple jobs and each jobs amount

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and customer, job, invoice; no operational business values.

Execution steps: Set up/use fixture helpers: customer, job, invoice.

Expected: One invoice retains multiple jobs and each jobs amount. Assert/check operations: assertEqualsCanonicalizing, assertSame.

Actual: Passed; 3 assertions; 0.042527 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessRelationshipsTest.php:102; Tests\Feature\Database\BusinessRelationshipsTest::test_one_invoice_retains_multiple_jobs_and_each_jobs_amount; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-086 - Payments allocations and reversals retain their relationships

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and customer, invoice, payment, code; no operational business values.

Execution steps: Set up/use fixture helpers: customer, invoice, payment, code.

Expected: Payments allocations and reversals retain their relationships. Assert/check operations: assertTrue, assertSame.

Actual: Passed; 7 assertions; 0.052255 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessRelationshipsTest.php:123; Tests\Feature\Database\BusinessRelationshipsTest::test_payments_allocations_and_reversals_retain_their_relationships; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-087 - Suppliers purchases and goods received notes share the ordered parts

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and code, part, location, employee; no operational business values.

Execution steps: Set up/use fixture helpers: code, part, location, employee.

Expected: Suppliers purchases and goods received notes share the ordered parts. Assert/check operations: assertTrue, assertSame.

Actual: Passed; 7 assertions; 0.070930 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessRelationshipsTest.php:161; Tests\Feature\Database\BusinessRelationshipsTest::test_suppliers_purchases_and_goods_received_notes_share_the_ordered_parts; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-088 - Pos and repair usage reference the same spare parts and movement history

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and part, location, movement, job, code, employee; no operational business values.

Execution steps: Set up/use fixture helpers: part, location, movement, job, code, employee.

Expected: Pos and repair usage reference the same spare parts and movement history. Assert/check operations: assertEqualsCanonicalizing, assertTrue.

Actual: Passed; 5 assertions; 0.075812 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessRelationshipsTest.php:210; Tests\Feature\Database\BusinessRelationshipsTest::test_pos_and_repair_usage_reference_the_same_spare_parts_and_movement_history; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-089 - An employee can link to one existing login account

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and employee; no operational business values.

Execution steps: Set up/use fixture helpers: employee.

Expected: An employee can link to one existing login account. Assert/check operations: assertTrue, assertArrayNotHasKey, assertDatabaseRejects.

Actual: Passed; 4 assertions; 0.050029 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessRelationshipsTest.php:259; Tests\Feature\Database\BusinessRelationshipsTest::test_an_employee_can_link_to_one_existing_login_account; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-090 - Money unit prices and quantities are returned as decimal strings

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and customer, invoice, payment, part, movement; no operational business values.

Execution steps: Set up/use fixture helpers: customer, invoice, payment, part, movement.

Expected: Money unit prices and quantities are returned as decimal strings. Assert/check operations: assertSame.

Actual: Passed; 6 assertions; 0.053512 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/BusinessRelationshipsTest.php:270; Tests\Feature\Database\BusinessRelationshipsTest::test_money_unit_prices_and_quantities_are_returned_as_decimal_strings; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-091 - The complete schema can migrate rollback and migrate again on an isolated database

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: The complete schema can migrate rollback and migrate again on an isolated database. Assert/check operations: assertSame, assertTrue, assertFalse.

Actual: Passed; 211 assertions; 1.149550 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/MigrationLifecycleTest.php:12; Tests\Feature\Database\MigrationLifecycleTest::test_the_complete_schema_can_migrate_rollback_and_migrate_again_on_an_isolated_database; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-092 - Foreign key enforcement is enabled on the test connection

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Foreign key enforcement is enabled on the test connection. Assert/check operations: assertSame.

Actual: Passed; 1 assertions; 0.038522 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/SchemaIntegrityTest.php:14; Tests\Feature\Database\SchemaIntegrityTest::test_foreign_key_enforcement_is_enabled_on_the_test_connection; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-093 - Transaction records are not cascade deleted

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Transaction records are not cascade deleted. Assert/check operations: assertNotEmpty, assertNotSame.

Actual: Passed; 86 assertions; 0.059146 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/SchemaIntegrityTest.php:25; Tests\Feature\Database\SchemaIntegrityTest::test_transaction_records_are_not_cascade_deleted; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-094 - No schema column uses binary floating point storage

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: No schema column uses binary floating point storage. Assert/check operations: assertNotContains.

Actual: Passed; 1672 assertions; 0.727206 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/SchemaIntegrityTest.php:44; Tests\Feature\Database\SchemaIntegrityTest::test_no_schema_column_uses_binary_floating_point_storage; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-095 - Mysql tables use innodb and utf8mb4

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and markTestSkipped; no operational business values.

Execution steps: Set up/use fixture helpers: markTestSkipped.

Expected: Mysql tables use innodb and utf8mb4. Assert/check operations: assertNotEmpty, assertSame, assertStringStartsWith.

Actual: Passed; 139 assertions; 0.034253 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/SchemaIntegrityTest.php:53; Tests\Feature\Database\SchemaIntegrityTest::test_mysql_tables_use_innodb_and_utf8mb4; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-096 - Mysql money fields are fixed precision decimals

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and markTestSkipped; no operational business values.

Execution steps: Set up/use fixture helpers: markTestSkipped.

Expected: Mysql money fields are fixed precision decimals. Assert/check operations: assertNotNull, assertSame, assertGreaterThanOrEqual.

Actual: Passed; 30 assertions; 0.061312 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/SchemaIntegrityTest.php:73; Tests\Feature\Database\SchemaIntegrityTest::test_mysql_money_fields_are_fixed_precision_decimals; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-097 - Mysql round trips amounts above binary floating point integer precision

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and markTestSkipped, customer; no operational business values.

Execution steps: Set up/use fixture helpers: markTestSkipped, customer.

Expected: Mysql round trips amounts above binary floating point integer precision. Assert/check operations: assertSame.

Actual: Passed; 1 assertions; 0.043624 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Database/SchemaIntegrityTest.php:101; Tests\Feature\Database\SchemaIntegrityTest::test_mysql_round_trips_amounts_above_binary_floating_point_integer_precision; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-098 - Administrator can register employee and link a hashed login account

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator; no operational business values.

Execution steps: Set up/use fixture helpers: administrator. Execute requests in method order: POST employees.store.

Expected: Administrator can register employee and link a hashed login account. Assert/check operations: assertRedirect, assertStringStartsWith, assertSame, assertTrue, assertDatabaseHas.

Actual: Passed; 7 assertions; 1.382553 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/EmployeeManagementTest.php:32; Tests\Feature\EmployeeManagementTest::test_administrator_can_register_employee_and_link_a_hashed_login_account; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-099 - Receptionist cannot register employees or create accounts

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute requests in method order: GET employees.create; POST employees.store.

Expected: Receptionist cannot register employees or create accounts. Assert/check operations: assertForbidden, assertDatabaseCount.

Actual: Passed; 3 assertions; 0.802782 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/EmployeeManagementTest.php:63; Tests\Feature\EmployeeManagementTest::test_receptionist_cannot_register_employees_or_create_accounts; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-100 - Username uniqueness is case insensitive and rejected before creation

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator; no operational business values.

Execution steps: Set up/use fixture helpers: administrator. Execute requests in method order: POST employees.store.

Expected: Username uniqueness is case insensitive and rejected before creation. Assert/check operations: assertRedirect, assertSessionHasErrors, assertDatabaseCount.

Actual: Passed; 5 assertions; 0.824484 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/EmployeeManagementTest.php:77; Tests\Feature\EmployeeManagementTest::test_username_uniqueness_is_case_insensitive_and_rejected_before_creation; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-101 - Employee ids are unique for multiple registrations

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator; no operational business values.

Execution steps: Set up/use fixture helpers: administrator.

Expected: Employee ids are unique for multiple registrations. Assert/check operations: assertNotSame, assertSame.

Actual: Passed; 2 assertions; 0.770581 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/EmployeeManagementTest.php:95; Tests\Feature\EmployeeManagementTest::test_employee_ids_are_unique_for_multiple_registrations; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-102 - Individual denial overrides role permission

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator; no operational business values.

Execution steps: Set up/use fixture helpers: administrator. Execute requests in method order: PUT admin.users.permissions; GET dashboard.

Expected: Individual denial overrides role permission. Assert/check operations: assertRedirect, assertForbidden.

Actual: Passed; 2 assertions; 0.833962 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/EmployeeManagementTest.php:107; Tests\Feature\EmployeeManagementTest::test_individual_denial_overrides_role_permission; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-103 - Administrator management pages render

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator, employee; no operational business values.

Execution steps: Set up/use fixture helpers: administrator, employee. Execute requests in method order: GET employees.index; GET employees.create; GET employees.show; GET admin.users.index; GET admin.users.show; GET admin.roles.index; GET admin.roles.edit; GET admin.permissions.index.

Expected: Administrator management pages render. Assert/check operations: assertOk.

Actual: Passed; 8 assertions; 1.604143 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/EmployeeManagementTest.php:121; Tests\Feature\EmployeeManagementTest::test_administrator_management_pages_render; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-104 - Technician policy only allows assigned jobs

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and employee, job; no operational business values.

Execution steps: Set up/use fixture helpers: employee, job.

Expected: Technician policy only allows assigned jobs. Assert/check operations: assertTrue, assertFalse.

Actual: Passed; 2 assertions; 0.857284 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/EmployeeManagementTest.php:138; Tests\Feature\EmployeeManagementTest::test_technician_policy_only_allows_assigned_jobs; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-105 - Employee deactivation disables linked login without deleting history

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator, employee; no operational business values.

Execution steps: Set up/use fixture helpers: administrator, employee. Execute requests in method order: PUT employees.update.

Expected: Employee deactivation disables linked login without deleting history. Assert/check operations: assertRedirect, assertDatabaseHas.

Actual: Passed; 5 assertions; 0.857424 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/EmployeeManagementTest.php:152; Tests\Feature\EmployeeManagementTest::test_employee_deactivation_disables_linked_login_without_deleting_history; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-106 - Failed employee account transaction leaves no partial records

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator, fail; no operational business values.

Execution steps: Set up/use fixture helpers: administrator, fail. Invoke services: EmployeeRegistrationService::register.

Expected: Failed employee account transaction leaves no partial records. Assert/check operations: assertDatabaseCount.

Actual: Passed; 2 assertions; 1.071159 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/EmployeeManagementTest.php:168; Tests\Feature\EmployeeManagementTest::test_failed_employee_account_transaction_leaves_no_partial_records; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-107 - Role seeder is idempotent and administrator receives all permissions

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed; no operational business values.

Execution steps: Set up/use fixture helpers: seed.

Expected: Role seeder is idempotent and administrator receives all permissions. Assert/check operations: assertSame.

Actual: Passed; 4 assertions; 1.669072 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/EmployeeManagementTest.php:189; Tests\Feature\EmployeeManagementTest::test_role_seeder_is_idempotent_and_administrator_receives_all_permissions; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-108 - Default role permissions follow the restricted matrix

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Default role permissions follow the restricted matrix. Assert/check operations: assertSame.

Actual: Passed; 20 assertions; 0.869490 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/EmployeeManagementTest.php:203; Tests\Feature\EmployeeManagementTest::test_default_role_permissions_follow_the_restricted_matrix; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-109 - Disabled user is logged out on the next request

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator; no operational business values.

Execution steps: Set up/use fixture helpers: administrator. Execute requests in method order: GET dashboard. Invoke services: UserAccountService::setEnabled.

Expected: Disabled user is logged out on the next request. Assert/check operations: assertRedirect, assertGuest.

Actual: Passed; 3 assertions; 0.774544 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/EmployeeManagementTest.php:248; Tests\Feature\EmployeeManagementTest::test_disabled_user_is_logged_out_on_the_next_request; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-110 - Last active administrator cannot be disabled

Module: Employees | BRD section(s): 14 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator, fail; no operational business values.

Execution steps: Set up/use fixture helpers: administrator, fail. Invoke services: UserAccountService::setEnabled.

Expected: Last active administrator cannot be disabled. Assert/check operations: assertArrayHasKey, assertTrue.

Actual: Passed; 2 assertions; 0.788698 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/EmployeeManagementTest.php:262; Tests\Feature\EmployeeManagementTest::test_last_active_administrator_cannot_be_disabled; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-111 - Creates unique draft invoices with multiple job totals

Module: Invoices | BRD section(s): 9, 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, billableJob, invoiceData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, billableJob, invoiceData. Execute requests in method order: POST invoices.store; POST invoices.store.

Expected: Creates unique draft invoices with multiple job totals. Assert/check operations: assertRedirect, assertSessionHasNoErrors, assertMatchesRegularExpression, assertSame, assertDatabaseCount, assertNotSame.

Actual: Passed; 14 assertions; 0.885034 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/InvoiceManagementTest.php:41; Tests\Feature\InvoiceManagementTest::test_creates_unique_draft_invoices_with_multiple_job_totals; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-112 - Invoice form restores customer and jobs after validation failure

Module: Invoices | BRD section(s): 9, 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, billableJob, invoiceData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, billableJob, invoiceData. Execute requests in method order: POST invoices.store; GET invoices.create.

Expected: Invoice form restores customer and jobs after validation failure. Assert/check operations: assertRedirect, assertSessionHasErrors, assertOk, assertSee.

Actual: Passed; 7 assertions; 1.065457 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/InvoiceManagementTest.php:74; Tests\Feature\InvoiceManagementTest::test_invoice_form_restores_customer_and_jobs_after_validation_failure; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-113 - Rejects cross customer duplicate and uncompleted jobs atomically

Module: Invoices | BRD section(s): 9, 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, billableJob, invoiceData, job; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, billableJob, invoiceData, job. Execute requests in method order: POST invoices.store; POST invoices.store; POST invoices.store.

Expected: Rejects cross customer duplicate and uncompleted jobs atomically. Assert/check operations: assertRedirect, assertSessionHasErrors, assertDatabaseCount.

Actual: Passed; 16 assertions; 0.871522 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/InvoiceManagementTest.php:91; Tests\Feature\InvoiceManagementTest::test_rejects_cross_customer_duplicate_and_uncompleted_jobs_atomically; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-114 - Actual issued part usages are itemized with decimal rounding without stock changes

Module: Invoices | BRD section(s): 9, 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, part, location, billableJob, code; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, part, location, billableJob, code, createIssuedInvoice.

Expected: Actual issued part usages are itemized with decimal rounding without stock changes. Assert/check operations: assertSame, assertDatabaseCount.

Actual: Passed; 6 assertions; 0.941506 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/InvoiceManagementTest.php:122; Tests\Feature\InvoiceManagementTest::test_actual_issued_part_usages_are_itemized_with_decimal_rounding_without_stock_changes; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-115 - Draft update appends jobs and saves delivery metadata without removing links

Module: Invoices | BRD section(s): 9, 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, billableJob, invoiceData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, billableJob, invoiceData. Execute requests in method order: POST invoices.store; PUT invoices.update.

Expected: Draft update appends jobs and saves delivery metadata without removing links. Assert/check operations: assertRedirect, assertSessionHasNoErrors, assertSame, assertDatabaseHas.

Actual: Passed; 8 assertions; 0.891554 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/InvoiceManagementTest.php:150; Tests\Feature\InvoiceManagementTest::test_draft_update_appends_jobs_and_saves_delivery_metadata_without_removing_links; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-116 - Invoice transaction rolls back when a billing line cannot be written

Module: Invoices | BRD section(s): 9, 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, billableJob, invoiceData, fail; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, billableJob, invoiceData, fail. Invoke services: InvoiceService::create.

Expected: Invoice transaction rolls back when a billing line cannot be written. Assert/check operations: assertDatabaseCount.

Actual: Passed; 3 assertions; 0.848815 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/InvoiceManagementTest.php:174; Tests\Feature\InvoiceManagementTest::test_invoice_transaction_rolls_back_when_a_billing_line_cannot_be_written; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-117 - Approved quotation prices are snapshotted and mismatches are blocked

Module: Invoices | BRD section(s): 9, 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, billableJob, approvedQuotation, invoiceData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, billableJob, approvedQuotation, invoiceData. Execute requests in method order: POST invoices.store; POST invoices.store.

Expected: Approved quotation prices are snapshotted and mismatches are blocked. Assert/check operations: assertRedirect, assertSessionHasNoErrors, assertSame, assertSessionHasErrors.

Actual: Passed; 12 assertions; 0.983658 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/InvoiceManagementTest.php:199; Tests\Feature\InvoiceManagementTest::test_approved_quotation_prices_are_snapshotted_and_mismatches_are_blocked; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-118 - Invoice balance uses existing advance and payment allocations without accepting payments

Module: Invoices | BRD section(s): 9, 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, billableJob, createIssuedInvoice, payment, allocation; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, billableJob, createIssuedInvoice, payment, allocation, code. Execute requests in method order: GET invoices.show. Invoke services: InvoiceBalanceService::summary.

Expected: Invoice balance uses existing advance and payment allocations without accepting payments. Assert/check operations: assertOk, assertSee, assertSame.

Actual: Passed; 11 assertions; 1.058899 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/InvoiceManagementTest.php:227; Tests\Feature\InvoiceManagementTest::test_invoice_balance_uses_existing_advance_and_payment_allocations_without_accepting_payments; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-119 - Issued snapshot is immutable and payment prevents cancellation

Module: Invoices | BRD section(s): 9, 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, billableJob, createIssuedInvoice, expectException; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, billableJob, createIssuedInvoice, expectException.

Expected: Issued snapshot is immutable and payment prevents cancellation. Assert/check operations: expectException, assertSame.

Actual: Passed; 3 assertions; 0.872100 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/InvoiceManagementTest.php:265; Tests\Feature\InvoiceManagementTest::test_issued_snapshot_is_immutable_and_payment_prevents_cancellation; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-120 - Invoice with allocated payment cannot be cancelled

Module: Invoices | BRD section(s): 9, 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, billableJob, createIssuedInvoice, payment, allocation; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, billableJob, createIssuedInvoice, payment, allocation. Execute requests in method order: POST invoices.cancel.

Expected: Invoice with allocated payment cannot be cancelled. Assert/check operations: assertRedirect, assertSessionHasErrors, assertSame.

Actual: Passed; 7 assertions; 1.018045 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/InvoiceManagementTest.php:279; Tests\Feature\InvoiceManagementTest::test_invoice_with_allocated_payment_cannot_be_cancelled; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-121 - Cancelled invoice keeps its number and job reservation

Module: Invoices | BRD section(s): 9, 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, billableJob, createIssuedInvoice, invoiceData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, billableJob, createIssuedInvoice, invoiceData. Execute requests in method order: POST invoices.cancel; POST invoices.store.

Expected: Cancelled invoice keeps its number and job reservation. Assert/check operations: assertRedirect, assertSessionHasNoErrors, assertSame, assertSessionHasErrors.

Actual: Passed; 12 assertions; 1.109818 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/InvoiceManagementTest.php:295; Tests\Feature\InvoiceManagementTest::test_cancelled_invoice_keeps_its_number_and_job_reservation; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-122 - Cashier can create issue and print but technician cannot manage invoices

Module: Invoices | BRD section(s): 9, 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, billableJob, invoiceData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, billableJob, invoiceData. Execute requests in method order: POST invoices.store; GET invoices.print; GET invoices.index; GET invoices.show.

Expected: Cashier can create issue and print but technician cannot manage invoices. Assert/check operations: assertRedirect, assertSessionHasNoErrors, assertOk, assertForbidden.

Actual: Passed; 5 assertions; 1.122018 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/InvoiceManagementTest.php:315; Tests\Feature\InvoiceManagementTest::test_cashier_can_create_issue_and_print_but_technician_cannot_manage_invoices; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-123 - Print duplicate and a5 pdf use saved data and log each output

Module: Invoices | BRD section(s): 9, 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, billableJob, createIssuedInvoice; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, billableJob, createIssuedInvoice. Execute requests in method order: GET invoices.print; GET invoices.duplicate; GET invoices.pdf.

Expected: Print duplicate and a5 pdf use saved data and log each output. Assert/check operations: assertOk, assertSee, assertHeader, assertStringStartsWith, assertStringContainsString, assertSame.

Actual: Passed; 15 assertions; 2.488330 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/InvoiceManagementTest.php:332; Tests\Feature\InvoiceManagementTest::test_print_duplicate_and_a5_pdf_use_saved_data_and_log_each_output; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-124 - Invoice print uses active template layout and footer

Module: Invoices | BRD section(s): 9, 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, billableJob, createIssuedInvoice; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, billableJob, createIssuedInvoice. Execute requests in method order: GET invoices.print.

Expected: Invoice print uses active template layout and footer. Assert/check operations: assertOk, assertSee, assertSame.

Actual: Passed; 9 assertions; 1.097324 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/InvoiceManagementTest.php:358; Tests\Feature\InvoiceManagementTest::test_invoice_print_uses_active_template_layout_and_footer; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-125 - Search and filters use invoice customer and repair job records

Module: Invoices | BRD section(s): 9, 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, billableJob, createIssuedInvoice; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, billableJob, createIssuedInvoice. Execute requests in method order: GET invoices.index; GET invoices.index.

Expected: Search and filters use invoice customer and repair job records. Assert/check operations: assertOk, assertSee.

Actual: Passed; 6 assertions; 1.227307 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/InvoiceManagementTest.php:394; Tests\Feature\InvoiceManagementTest::test_search_and_filters_use_invoice_customer_and_repair_job_records; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-126 - Step four customer search and billable job lookup are reused

Module: Invoices | BRD section(s): 9, 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, billableJob; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, billableJob. Execute requests in method order: GET customers.search; GET invoices.jobs.

Expected: Step four customer search and billable job lookup are reused. Assert/check operations: assertOk, assertJsonPath.

Actual: Passed; 4 assertions; 0.901569 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/InvoiceManagementTest.php:409; Tests\Feature\InvoiceManagementTest::test_step_four_customer_search_and_billable_job_lookup_are_reused; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-127 - Correcting a payment changes collections without reducing the issued sale

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed, customer, invoice, payment; no operational business values.

Execution steps: Set up/use fixture helpers: seed, customer, invoice, payment.

Expected: Correcting a payment changes collections without reducing the issued sale. Assert/check operations: assertSame.

Actual: Passed; 2 assertions; 0.805557 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/LatestPurchaseCostTest.php:24; Tests\Feature\LatestPurchaseCostTest::test_correcting_a_payment_changes_collections_without_reducing_the_issued_sale; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-128 - Old customer return does not replace latest purchase cost for pos or valuation

Module: Inventory | BRD section(s): 12 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed, employee, location, part, code; no operational business values.

Execution steps: Set up/use fixture helpers: seed, employee, location, part, code. Execute requests in method order: GET reports.show. Invoke services: POSCheckoutService::checkout, POSReturnService::record, InventoryService::latestUnitCost, ReportDataService::build.

Expected: Old customer return does not replace latest purchase cost for pos or valuation. Assert/check operations: assertSame, assertOk.

Actual: Passed; 8 assertions; 1.180039 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/LatestPurchaseCostTest.php:40; Tests\Feature\LatestPurchaseCostTest::test_old_customer_return_does_not_replace_latest_purchase_cost_for_pos_or_valuation; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-129 - Cashier scans repeated barcodes and completes sale with shared stock and payment

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart. Execute requests in method order: GET pos.index; POST pos.cart.add; POST pos.cart.add; POST pos.checkout; GET pos.sales.show; GET pos.sales.print; GET pos.sales.reprint. Invoke services: InventoryService::availableAtDefaultLocation.

Expected: Cashier scans repeated barcodes and completes sale with shared stock and payment. Assert/check operations: assertOk, assertSee, assertRedirect, assertSessionHasNoErrors, assertSame, assertDatabaseCount, assertDatabaseHas.

Actual: Passed; 25 assertions; 1.210574 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:32; Tests\Feature\POSManagementTest::test_cashier_scans_repeated_barcodes_and_completes_sale_with_shared_stock_and_payment; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-130 - Unverified bank and cheque sales do not post payment or reduce stock until authorized

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart. Execute requests in method order: POST pos.cart.add; POST pos.checkout; POST pos.sales.verify.

Expected: Unverified bank and cheque sales do not post payment or reduce stock until authorized. Assert/check operations: assertRedirect, assertSame, assertDatabaseCount.

Actual: Passed; 11 assertions; 0.886439 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:77; Tests\Feature\POSManagementTest::test_unverified_bank_and_cheque_sales_do_not_post_payment_or_reduce_stock_until_authorized; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-131 - Checkout revalidates stock and idempotency prevents duplicate cash postings

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart. Execute requests in method order: POST pos.checkout; POST pos.checkout; POST pos.checkout; POST pos.checkout.

Expected: Checkout revalidates stock and idempotency prevents duplicate cash postings. Assert/check operations: assertSessionHasErrors, assertDatabaseCount, assertRedirect.

Actual: Passed; 11 assertions; 0.925384 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:105; Tests\Feature\POSManagementTest::test_checkout_revalidates_stock_and_idempotency_prevents_duplicate_cash_postings; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-132 - Pos payments appear in payment history and show their sale reference

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart. Execute requests in method order: POST pos.checkout; GET payments.index; GET payments.show; GET payments.print.

Expected: Pos payments appear in payment history and show their sale reference. Assert/check operations: assertRedirect, assertOk, assertSee.

Actual: Passed; 9 assertions; 1.139614 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:144; Tests\Feature\POSManagementTest::test_pos_payments_appear_in_payment_history_and_show_their_sale_reference; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-133 - Cashier cannot apply discount or view pos without permission

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart. Execute requests in method order: POST pos.checkout.

Expected: Cashier cannot apply discount or view pos without permission. Assert/check operations: assertForbidden, assertDatabaseCount.

Actual: Passed; 2 assertions; 0.881726 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:167; Tests\Feature\POSManagementTest::test_cashier_cannot_apply_discount_or_view_pos_without_permission; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-134 - Returns preserve original sale and restore only resalable stock with authorized refunds

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart. Execute requests in method order: POST pos.checkout; POST pos.returns.store; POST pos.returns.store; POST pos.returns.store; POST pos.returns.store; GET pos.reports; POST pos.returns.store.

Expected: Returns preserve original sale and restore only resalable stock with authorized refunds. Assert/check operations: assertRedirect, assertSessionHasErrors, assertDatabaseCount, assertSame, assertDatabaseHas, assertOk, assertSee.

Actual: Passed; 17 assertions; 1.021911 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:181; Tests\Feature\POSManagementTest::test_returns_preserve_original_sale_and_restore_only_resalable_stock_with_authorized_refunds; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-135 - Cashier may not record a pos return or manager verification without role permission

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart. Execute requests in method order: GET pos.returns.index; POST pos.returns.store.

Expected: Cashier may not record a pos return or manager verification without role permission. Assert/check operations: assertForbidden.

Actual: Passed; 2 assertions; 0.837663 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:238; Tests\Feature\POSManagementTest::test_cashier_may_not_record_a_pos_return_or_manager_verification_without_role_permission; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-136 - Whole unit products reject fractional cart checkout and returns without server errors

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart, cashCheckoutData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart, cashCheckoutData. Execute requests in method order: POST pos.cart.add; PUT pos.cart.update; POST pos.checkout; POST pos.checkout; POST pos.returns.store.

Expected: Whole unit products reject fractional cart checkout and returns without server errors. Assert/check operations: assertSessionHasErrors, assertDatabaseCount, assertRedirect, assertSessionHasNoErrors.

Actual: Passed; 12 assertions; 0.909450 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:246; Tests\Feature\POSManagementTest::test_whole_unit_products_reject_fractional_cart_checkout_and_returns_without_server_errors; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-137 - Discounted returns cannot refund the undiscounted item value

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart, cashCheckoutData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart, cashCheckoutData. Execute requests in method order: POST pos.checkout; POST pos.returns.store; POST pos.returns.store.

Expected: Discounted returns cannot refund the undiscounted item value. Assert/check operations: assertSessionHasNoErrors, assertSessionHasErrors, assertDatabaseCount, assertDatabaseHas.

Actual: Passed; 6 assertions; 0.897843 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:269; Tests\Feature\POSManagementTest::test_discounted_returns_cannot_refund_the_undiscounted_item_value; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-138 - Receipt uses sale snapshots and all paper formats generate pdfs

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart, cashCheckoutData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart, cashCheckoutData. Execute requests in method order: POST pos.checkout; GET pos.sales.print; GET pos.sales.pdf.

Expected: Receipt uses sale snapshots and all paper formats generate pdfs. Assert/check operations: assertSessionHasNoErrors, assertOk, assertSee, assertDontSee, assertHeader, assertStringStartsWith, assertDatabaseCount.

Actual: Passed; 19 assertions; 4.326011 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:290; Tests\Feature\POSManagementTest::test_receipt_uses_sale_snapshots_and_all_paper_formats_generate_pdfs; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-139 - Pending sale cancellation has no stock or payment effects and cannot be verified

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart. Execute requests in method order: POST pos.checkout; GET pos.sales.print; POST pos.sales.cancel; POST pos.sales.verify.

Expected: Pending sale cancellation has no stock or payment effects and cannot be verified. Assert/check operations: assertSessionHasNoErrors, assertNotFound, assertSessionHasErrors, assertDatabaseCount.

Actual: Passed; 6 assertions; 0.893941 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:310; Tests\Feature\POSManagementTest::test_pending_sale_cancellation_has_no_stock_or_payment_effects_and_cannot_be_verified; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-140 - Client totals are ignored and inactive customers are rejected

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart, customer, cashCheckoutData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart, customer, cashCheckoutData. Execute requests in method order: POST pos.checkout; POST pos.checkout.

Expected: Client totals are ignored and inactive customers are rejected. Assert/check operations: assertSessionHasErrors, assertDatabaseCount, assertRedirect, assertSessionHasNoErrors, assertDatabaseHas.

Actual: Passed; 6 assertions; 0.850381 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:329; Tests\Feature\POSManagementTest::test_client_totals_are_ignored_and_inactive_customers_are_rejected; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-141 - Cart refresh update remove and clear never change inventory

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart. Execute requests in method order: POST pos.cart.add; GET pos.index; PUT pos.cart.update; DELETE pos.cart.remove; POST pos.cart.add; DELETE pos.cart.clear.

Expected: Cart refresh update remove and clear never change inventory. Assert/check operations: assertRedirect, assertOk, assertSame, assertSessionHasNoErrors, assertNull, assertDatabaseCount.

Actual: Passed; 13 assertions; 0.957382 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:343; Tests\Feature\POSManagementTest::test_cart_refresh_update_remove_and_clear_never_change_inventory; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-142 - Stock posting failure rolls back sale payment allocation and number

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart, partialMock, cashCheckoutData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart, partialMock, cashCheckoutData. Execute requests in method order: POST pos.checkout.

Expected: Stock posting failure rolls back sale payment allocation and number. Assert/check operations: assertStatus, assertDatabaseCount, assertDatabaseHas.

Actual: Passed; 8 assertions; 2.022293 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:362; Tests\Feature\POSManagementTest::test_stock_posting_failure_rolls_back_sale_payment_allocation_and_number; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-143 - Refund only reporting period includes negative net sales and collections

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart, travelTo, cashCheckoutData, travelBack; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart, travelTo, cashCheckoutData, travelBack. Execute requests in method order: POST pos.checkout; POST pos.returns.store; GET pos.reports.

Expected: Refund only reporting period includes negative net sales and collections. Assert/check operations: assertSessionHasNoErrors, assertOk, assertViewHas.

Actual: Passed; 5 assertions; 1.114638 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:379; Tests\Feature\POSManagementTest::test_refund_only_reporting_period_includes_negative_net_sales_and_collections; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-144 - Final partial return can refund the remaining rounding cent

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart, cashCheckoutData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart, cashCheckoutData. Execute requests in method order: POST pos.checkout; POST pos.returns.store. Invoke services: InventoryService::availableAtDefaultLocation.

Expected: Final partial return can refund the remaining rounding cent. Assert/check operations: assertSessionHasNoErrors, assertSame.

Actual: Passed; 7 assertions; 1.200712 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:399; Tests\Feature\POSManagementTest::test_final_partial_return_can_refund_the_remaining_rounding_cent; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-145 - Discount rounding across lines allows exact full refund

Module: POS | BRD section(s): 12, 14, 23 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, saleablePart, cashCheckoutData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, saleablePart, cashCheckoutData. Execute requests in method order: POST pos.checkout; POST pos.returns.store.

Expected: Discount rounding across lines allows exact full refund. Assert/check operations: assertSessionHasNoErrors, assertSame.

Actual: Passed; 5 assertions; 1.027448 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/POSManagementTest.php:419; Tests\Feature\POSManagementTest::test_discount_rounding_across_lines_allows_exact_full_refund; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-146 - Cash bank and cheque partial payments settle an invoice

Module: Payments | BRD section(s): 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, issuedInvoice, postPayment; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, issuedInvoice, postPayment. Execute requests in method order: GET payments.index; GET credit-invoices.index. Invoke services: InvoiceBalanceService::summary.

Expected: Cash bank and cheque partial payments settle an invoice. Assert/check operations: assertRedirect, assertSessionHasNoErrors, assertSame, assertDatabaseCount, assertDatabaseHas, assertOk, assertSee.

Actual: Passed; 18 assertions; 1.140222 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/PaymentManagementTest.php:34; Tests\Feature\PaymentManagementTest::test_cash_bank_and_cheque_partial_payments_settle_an_invoice; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-147 - Customer advance is applied once when invoice is issued

Module: Payments | BRD section(s): 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, issuedInvoice; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, issuedInvoice. Invoke services: PaymentService::receiveAdvance, InvoiceBalanceService::summary, PaymentAllocationService::applyAvailableAdvances.

Expected: Customer advance is applied once when invoice is issued. Assert/check operations: assertSame, assertDatabaseCount, assertDatabaseHas.

Actual: Passed; 9 assertions; 0.926142 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/PaymentManagementTest.php:65; Tests\Feature\PaymentManagementTest::test_customer_advance_is_applied_once_when_invoice_is_issued; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-148 - Job advance is linked to the customer job and printed receipt

Module: Payments | BRD section(s): 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, job; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, job. Execute requests in method order: GET payments.print. Invoke services: PaymentService::receiveAdvance.

Expected: Job advance is linked to the customer job and printed receipt. Assert/check operations: assertSame, assertOk, assertSee.

Actual: Passed; 5 assertions; 0.895903 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/PaymentManagementTest.php:94; Tests\Feature\PaymentManagementTest::test_job_advance_is_linked_to_the_customer_job_and_printed_receipt; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-149 - Repair receipt records authorized advance and snapshots payment reference

Module: Payments | BRD section(s): 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer. Execute requests in method order: GET repair-receipts.show. Invoke services: RepairReceiptService::register.

Expected: Repair receipt records authorized advance and snapshots payment reference. Assert/check operations: assertSame, assertDatabaseHas, assertOk, assertSee.

Actual: Passed; 6 assertions; 1.007130 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/PaymentManagementTest.php:115; Tests\Feature\PaymentManagementTest::test_repair_receipt_records_authorized_advance_and_snapshots_payment_reference; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-150 - Idempotency blocks duplicate invoice payment and overpayment

Module: Payments | BRD section(s): 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, issuedInvoice; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, issuedInvoice. Execute requests in method order: POST payments.store; POST payments.store; POST payments.store.

Expected: Idempotency blocks duplicate invoice payment and overpayment. Assert/check operations: assertRedirect, assertSessionHasNoErrors, assertSame, assertDatabaseCount, assertSessionHasErrors.

Actual: Passed; 13 assertions; 1.101939 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/PaymentManagementTest.php:147; Tests\Feature\PaymentManagementTest::test_idempotency_blocks_duplicate_invoice_payment_and_overpayment; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-151 - Cross customer allocation is rejected without creating an allocation

Module: Payments | BRD section(s): 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, issuedInvoice, payment, fail; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, issuedInvoice, payment, fail. Invoke services: PaymentAllocationService::allocateToInvoice.

Expected: Cross customer allocation is rejected without creating an allocation. Assert/check operations: assertArrayHasKey, assertDatabaseCount.

Actual: Passed; 4 assertions; 0.960358 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/PaymentManagementTest.php:176; Tests\Feature\PaymentManagementTest::test_cross_customer_allocation_is_rejected_without_creating_an_allocation; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-152 - Refund and full accounting reversal restore invoice balance and log entries

Module: Payments | BRD section(s): 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, issuedInvoice, paymentData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, issuedInvoice, paymentData. Invoke services: PaymentService::receiveInvoicePayment, PaymentService::reverse, InvoiceBalanceService::summary.

Expected: Refund and full accounting reversal restore invoice balance and log entries. Assert/check operations: assertSame, assertDatabaseHas.

Actual: Passed; 8 assertions; 0.972056 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/PaymentManagementTest.php:203; Tests\Feature\PaymentManagementTest::test_refund_and_full_accounting_reversal_restore_invoice_balance_and_log_entries; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-153 - Only authorized roles can reverse and receiver impersonation is restricted

Module: Payments | BRD section(s): 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, issuedInvoice, paymentData, employee; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, issuedInvoice, paymentData, employee. Execute requests in method order: POST payments.reverse; POST payments.store. Invoke services: PaymentService::receiveInvoicePayment.

Expected: Only authorized roles can reverse and receiver impersonation is restricted. Assert/check operations: assertForbidden, assertRedirect, assertSessionHasErrors, assertDatabaseCount.

Actual: Passed; 8 assertions; 0.930851 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/PaymentManagementTest.php:237; Tests\Feature\PaymentManagementTest::test_only_authorized_roles_can_reverse_and_receiver_impersonation_is_restricted; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-154 - Customer statement uses ledger running balance and pdf route

Module: Payments | BRD section(s): 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, issuedInvoice, postPayment; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, issuedInvoice, postPayment. Execute requests in method order: GET customers.statement; GET customers.statement.print; GET customers.statement.pdf.

Expected: Customer statement uses ledger running balance and pdf route. Assert/check operations: assertRedirect, assertOk, assertSee, assertHeader.

Actual: Passed; 11 assertions; 2.201759 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/PaymentManagementTest.php:257; Tests\Feature\PaymentManagementTest::test_customer_statement_uses_ledger_running_balance_and_pdf_route; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-155 - Credit collections are permission checked and support repeated partial payments

Module: Payments | BRD section(s): 10, 11 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, issuedInvoice, postPayment; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, issuedInvoice, postPayment. Execute requests in method order: GET payments.create; GET credit-invoices.collect; GET credit-invoices.show; GET credit-invoices.index. Invoke services: InvoiceBalanceService::summary.

Expected: Credit collections are permission checked and support repeated partial payments. Assert/check operations: assertOk, assertSee, assertRedirect, assertSessionHasNoErrors, assertSame.

Actual: Passed; 16 assertions; 1.177669 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/PaymentManagementTest.php:284; Tests\Feature\PaymentManagementTest::test_credit_collections_are_permission_checked_and_support_repeated_partial_payments; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-156 - Customer statement uses template title paper orientation sinhala font and print history

Module: Templates | BRD section(s): 16, 19 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator, customer; no operational business values.

Execution steps: Set up/use fixture helpers: administrator, customer. Execute requests in method order: GET customers.statement.print; GET customers.statement.pdf.

Expected: Customer statement uses template title paper orientation sinhala font and print history. Assert/check operations: assertOk, assertSee, assertStringContainsString, assertDatabaseHas.

Actual: Passed; 7 assertions; 2.003710 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/PrintGapCompletionTest.php:29; Tests\Feature\PrintGapCompletionTest::test_customer_statement_uses_template_title_paper_orientation_sinhala_font_and_print_history; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-157 - Purchase and grn print pdf and duplicate output preserve their source

Module: Templates | BRD section(s): 16, 19 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator, code, location, part; no operational business values.

Execution steps: Set up/use fixture helpers: administrator, code, location, part. Execute requests in method order: GET purchases.document.

Expected: Purchase and grn print pdf and duplicate output preserve their source. Assert/check operations: assertOk, assertSee, assertHeader, assertStringStartsWith, assertDatabaseCount, assertDatabaseHas.

Actual: Passed; 14 assertions; 3.768965 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/PrintGapCompletionTest.php:42; Tests\Feature\PrintGapCompletionTest::test_purchase_and_grn_print_pdf_and_duplicate_output_preserve_their_source; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-158 - Print template edits create new active versions and can reset to original

Module: Templates | BRD section(s): 16, 19 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed; no operational business values.

Execution steps: Set up/use fixture helpers: seed. Execute requests in method order: GET print-templates.preview; PUT print-templates.update; POST print-templates.reset.

Expected: Print template edits create new active versions and can reset to original. Assert/check operations: assertOk, assertSee, assertRedirect, assertTrue, assertFalse, assertSame, assertDatabaseHas.

Actual: Passed; 11 assertions; 0.957256 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/PrintTemplateManagementTest.php:15; Tests\Feature\PrintTemplateManagementTest::test_print_template_edits_create_new_active_versions_and_can_reset_to_original; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-159 - Paper size validation matches quotation renderer support

Module: Templates | BRD section(s): 16, 19 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed; no operational business values.

Execution steps: Set up/use fixture helpers: seed. Execute requests in method order: PUT print-templates.update.

Expected: Paper size validation matches quotation renderer support. Assert/check operations: assertSessionHasErrors, assertSame.

Actual: Passed; 3 assertions; 0.860142 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/PrintTemplateManagementTest.php:67; Tests\Feature\PrintTemplateManagementTest::test_paper_size_validation_matches_quotation_renderer_support; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-160 - Draft quote calculates decimal totals without touching inventory or billing

Module: Quotations | BRD section(s): 8 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, job, quotationData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, job, quotationData. Execute requests in method order: POST quotations.store.

Expected: Draft quote calculates decimal totals without touching inventory or billing. Assert/check operations: assertRedirect, assertSessionHasNoErrors, assertMatchesRegularExpression, assertSame, assertDatabaseCount.

Actual: Passed; 12 assertions; 1.048850 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/QuotationManagementTest.php:32; Tests\Feature\QuotationManagementTest::test_draft_quote_calculates_decimal_totals_without_touching_inventory_or_billing; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-161 - Draft edits retain removed lines and audit the line level change

Module: Quotations | BRD section(s): 8 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, job, quotationData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, job, quotationData. Execute requests in method order: POST quotations.store; PUT quotations.update.

Expected: Draft edits retain removed lines and audit the line level change. Assert/check operations: assertRedirect, assertSessionHasNoErrors, assertSame, assertCount.

Actual: Passed; 11 assertions; 1.025217 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/QuotationManagementTest.php:55; Tests\Feature\QuotationManagementTest::test_draft_edits_retain_removed_lines_and_audit_the_line_level_change; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-162 - Issue and customer approval are recorded with job status history

Module: Quotations | BRD section(s): 8 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, job, quotationData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, job, quotationData. Execute requests in method order: POST quotations.store; POST quotations.decision.

Expected: Issue and customer approval are recorded with job status history. Assert/check operations: assertRedirect, assertSessionHasNoErrors, assertSame, assertNotNull, assertDatabaseCount.

Actual: Passed; 20 assertions; 0.984159 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/QuotationManagementTest.php:91; Tests\Feature\QuotationManagementTest::test_issue_and_customer_approval_are_recorded_with_job_status_history; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-163 - Customer rejection sets customer contact failed and is append only

Module: Quotations | BRD section(s): 8 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, job, issueQuotation; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, job, issueQuotation. Execute requests in method order: POST quotations.decision.

Expected: Customer rejection sets customer contact failed and is append only. Assert/check operations: assertRedirect, assertSessionHasNoErrors, assertSame, assertDatabaseHas.

Actual: Passed; 8 assertions; 0.881947 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/QuotationManagementTest.php:134; Tests\Feature\QuotationManagementTest::test_customer_rejection_sets_customer_contact_failed_and_is_append_only; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-164 - Revision copies quote items and supersedes the previous issued quote

Module: Quotations | BRD section(s): 8 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, job, issueQuotation; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, job, issueQuotation. Execute requests in method order: POST quotations.decision; POST quotations.revise; POST quotations.issue.

Expected: Revision copies quote items and supersedes the previous issued quote. Assert/check operations: assertRedirect, assertSessionHasNoErrors, assertSame.

Actual: Passed; 14 assertions; 0.952094 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/QuotationManagementTest.php:159; Tests\Feature\QuotationManagementTest::test_revision_copies_quote_items_and_supersedes_the_previous_issued_quote; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-165 - Revision of expired quotation gets a fresh validity period

Module: Quotations | BRD section(s): 8 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, job, issueQuotation, travel, travelBack; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, job, issueQuotation, travel, travelBack. Execute requests in method order: POST quotations.revise.

Expected: Revision of expired quotation gets a fresh validity period. Assert/check operations: assertRedirect, assertSessionHasNoErrors, assertSame.

Actual: Passed; 6 assertions; 0.961138 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/QuotationManagementTest.php:194; Tests\Feature\QuotationManagementTest::test_revision_of_expired_quotation_gets_a_fresh_validity_period; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-166 - Unapproved quotation is not visible to cashier

Module: Quotations | BRD section(s): 8 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, job, issueQuotation; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, job, issueQuotation. Execute requests in method order: GET quotations.show.

Expected: Unapproved quotation is not visible to cashier. Assert/check operations: assertForbidden.

Actual: Passed; 3 assertions; 0.912335 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/QuotationManagementTest.php:214; Tests\Feature\QuotationManagementTest::test_unapproved_quotation_is_not_visible_to_cashier; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-167 - Repair cannot start until an issued quotation is approved

Module: Quotations | BRD section(s): 8 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, job, issueQuotation, fail; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, job, issueQuotation, fail. Invoke services: RepairJobService::updateStatus.

Expected: Repair cannot start until an issued quotation is approved. Assert/check operations: assertArrayHasKey, assertSame.

Actual: Passed; 4 assertions; 0.989052 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/QuotationManagementTest.php:226; Tests\Feature\QuotationManagementTest::test_repair_cannot_start_until_an_issued_quotation_is_approved; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-168 - Expired quotation cannot be approved

Module: Quotations | BRD section(s): 8 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, job, issueQuotation, travel, travelBack; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, job, issueQuotation, travel, travelBack. Execute requests in method order: POST quotations.decision.

Expected: Expired quotation cannot be approved. Assert/check operations: assertRedirect, assertSessionHasErrors, assertSame.

Actual: Passed; 9 assertions; 0.867145 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/QuotationManagementTest.php:245; Tests\Feature\QuotationManagementTest::test_expired_quotation_cannot_be_approved; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-169 - Future customer decision date is rejected without saving a decision

Module: Quotations | BRD section(s): 8 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, job, issueQuotation; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, job, issueQuotation. Execute requests in method order: POST quotations.decision.

Expected: Future customer decision date is rejected without saving a decision. Assert/check operations: assertRedirect, assertSessionHasErrors, assertSame.

Actual: Passed; 9 assertions; 0.893924 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/QuotationManagementTest.php:269; Tests\Feature\QuotationManagementTest::test_future_customer_decision_date_is_rejected_without_saving_a_decision; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-170 - A4 and a5 pdfs render with sinhala font and log each export

Module: Quotations | BRD section(s): 8 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, job, issueQuotation; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, job, issueQuotation. Execute requests in method order: GET quotations.pdf; GET quotations.pdf.

Expected: A4 and a5 pdfs render with sinhala font and log each export. Assert/check operations: assertOk, assertHeader, assertStringStartsWith, assertStringContainsString, assertSame.

Actual: Passed; 12 assertions; 3.283098 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/QuotationManagementTest.php:291; Tests\Feature\QuotationManagementTest::test_a4_and_a5_pdfs_render_with_sinhala_font_and_log_each_export; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-171 - Receptionist saves receipt and creates linked job history and a6 card

Module: Receipts | BRD section(s): 5 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, customerAttributes, employee, receiptData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, customerAttributes, employee, receiptData. Execute requests in method order: POST repair-receipts.store.

Expected: Receptionist saves receipt and creates linked job history and a6 card. Assert/check operations: assertRedirect, assertSessionHasNoErrors, assertMatchesRegularExpression, assertSame, assertDatabaseHas.

Actual: Passed; 14 assertions; 0.943377 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/RepairReceiptAndJobManagementTest.php:34; Tests\Feature\RepairReceiptAndJobManagementTest::test_receptionist_saves_receipt_and_creates_linked_job_history_and_a6_card; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-172 - Automatic references use the database sequence and manual numbers are unique

Module: Receipts | BRD section(s): 5 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, employee, receiptData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, employee, receiptData. Execute requests in method order: POST repair-receipts.store; POST repair-receipts.store; POST repair-receipts.store.

Expected: Automatic references use the database sequence and manual numbers are unique. Assert/check operations: assertRedirect, assertCount, assertMatchesRegularExpression, assertSessionHasErrors, assertDatabaseCount.

Actual: Passed; 11 assertions; 0.945721 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/RepairReceiptAndJobManagementTest.php:74; Tests\Feature\RepairReceiptAndJobManagementTest::test_automatic_references_use_the_database_sequence_and_manual_numbers_are_unique; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-173 - Receipt validation rejects inactive customers and missing required fields

Module: Receipts | BRD section(s): 5 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, employee, receiptData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, employee, receiptData. Execute requests in method order: POST repair-receipts.store.

Expected: Receipt validation rejects inactive customers and missing required fields. Assert/check operations: assertRedirect, assertSessionHasErrors, assertDatabaseCount.

Actual: Passed; 9 assertions; 0.862622 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/RepairReceiptAndJobManagementTest.php:105; Tests\Feature\RepairReceiptAndJobManagementTest::test_receipt_validation_rejects_inactive_customers_and_missing_required_fields; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-174 - Advance payment is recorded once in payment allocation and customer ledger

Module: Receipts | BRD section(s): 5 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, employee, receiptData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, employee, receiptData. Execute requests in method order: POST repair-receipts.store.

Expected: Advance payment is recorded once in payment allocation and customer ledger. Assert/check operations: assertRedirect, assertSame.

Actual: Passed; 8 assertions; 0.897141 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/RepairReceiptAndJobManagementTest.php:125; Tests\Feature\RepairReceiptAndJobManagementTest::test_advance_payment_is_recorded_once_in_payment_allocation_and_customer_ledger; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-175 - Receipt and duplicate print use snapshot without job type and log each copy

Module: Receipts | BRD section(s): 5 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, employee, receiptData; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, employee, receiptData. Execute requests in method order: POST repair-receipts.store; GET repair-receipts.print; GET repair-receipts.duplicate.

Expected: Receipt and duplicate print use snapshot without job type and log each copy. Assert/check operations: assertRedirect, assertOk, assertSee, assertDontSee, assertDatabaseCount, assertDatabaseHas, assertSame.

Actual: Passed; 11 assertions; 0.913489 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/RepairReceiptAndJobManagementTest.php:152; Tests\Feature\RepairReceiptAndJobManagementTest::test_receipt_and_duplicate_print_use_snapshot_without_job_type_and_log_each_copy; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-176 - Failed advance recording rolls back the entire job registration

Module: Receipts | BRD section(s): 5 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, employee, receiptData, fail; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, employee, receiptData, fail. Invoke services: RepairReceiptService::register.

Expected: Failed advance recording rolls back the entire job registration. Assert/check operations: assertDatabaseCount.

Actual: Passed; 6 assertions; 0.903686 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/RepairReceiptAndJobManagementTest.php:182; Tests\Feature\RepairReceiptAndJobManagementTest::test_failed_advance_recording_rolls_back_the_entire_job_registration; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-177 - Authorized status changes append history and do not invent transitions

Module: Receipts | BRD section(s): 5 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, job; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, job. Execute requests in method order: POST repair-jobs.update-status.

Expected: Authorized status changes append history and do not invent transitions. Assert/check operations: assertRedirect, assertSame, assertDatabaseHas.

Actual: Passed; 5 assertions; 0.818159 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/RepairReceiptAndJobManagementTest.php:205; Tests\Feature\RepairReceiptAndJobManagementTest::test_authorized_status_changes_append_history_and_do_not_invent_transitions; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-178 - Job assignment and record level access are limited to active technicians

Module: Receipts | BRD section(s): 5 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, employee, job; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, employee, job. Execute requests in method order: POST repair-jobs.assign; POST repair-jobs.assign; GET repair-jobs.show; GET repair-jobs.show; GET repair-jobs.index.

Expected: Job assignment and record level access are limited to active technicians. Assert/check operations: assertRedirect, assertSame, assertDatabaseHas, assertSessionHasErrors, assertOk, assertForbidden, assertDontSee.

Actual: Passed; 11 assertions; 1.058539 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/RepairReceiptAndJobManagementTest.php:227; Tests\Feature\RepairReceiptAndJobManagementTest::test_job_assignment_and_record_level_access_are_limited_to_active_technicians; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-179 - Receptionist can register and print but cannot assign or change status

Module: Receipts | BRD section(s): 5 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, job, employee; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, job, employee. Execute requests in method order: GET repair-receipts.index; GET repair-jobs.status-search; POST repair-jobs.update-status; POST repair-jobs.assign.

Expected: Receptionist can register and print but cannot assign or change status. Assert/check operations: assertOk, assertForbidden.

Actual: Passed; 4 assertions; 1.009984 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/RepairReceiptAndJobManagementTest.php:255; Tests\Feature\RepairReceiptAndJobManagementTest::test_receptionist_can_register_and_print_but_cannot_assign_or_change_status; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-180 - Technician can record inspection and repair notes without stock deduction

Module: Receipts | BRD section(s): 5 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, employee, job; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, employee, job. Execute requests in method order: POST repair-jobs.inspect; POST repair-jobs.save-repair.

Expected: Technician can record inspection and repair notes without stock deduction. Assert/check operations: assertRedirect, assertSame, assertDatabaseHas, assertDatabaseCount.

Actual: Passed; 9 assertions; 0.832306 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/RepairReceiptAndJobManagementTest.php:270; Tests\Feature\RepairReceiptAndJobManagementTest::test_technician_can_record_inspection_and_repair_notes_without_stock_deduction; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-181 - Job card print is two page a6 and contains writable work areas

Module: Receipts | BRD section(s): 5 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, job; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, job. Execute requests in method order: GET job-cards.print.

Expected: Job card print is two page a6 and contains writable work areas. Assert/check operations: assertOk, assertSee, assertDatabaseHas.

Actual: Passed; 9 assertions; 0.919626 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/RepairReceiptAndJobManagementTest.php:303; Tests\Feature\RepairReceiptAndJobManagementTest::test_job_card_print_is_two_page_a6_and_contains_writable_work_areas; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-182 - Search and filters return database paginated jobs

Module: Receipts | BRD section(s): 5 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and userWithRole, customer, customerAttributes, job; no operational business values.

Execution steps: Set up/use fixture helpers: userWithRole, customer, customerAttributes, job. Execute requests in method order: GET repair-jobs.status-search; GET repair-receipts.index.

Expected: Search and filters return database paginated jobs. Assert/check operations: assertOk, assertSee, assertDontSee.

Actual: Passed; 7 assertions; 1.115948 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/RepairReceiptAndJobManagementTest.php:324; Tests\Feature\RepairReceiptAndJobManagementTest::test_search_and_filters_return_database_paginated_jobs; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-183 - Repair job identifiers cannot be changed after creation

Module: Receipts | BRD section(s): 5 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and job, expectException; no operational business values.

Execution steps: Set up/use fixture helpers: job, expectException.

Expected: Repair job identifiers cannot be changed after creation. Assert/check operations: expectException.

Actual: Passed; 1 assertions; 0.998204 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/RepairReceiptAndJobManagementTest.php:357; Tests\Feature\RepairReceiptAndJobManagementTest::test_repair_job_identifiers_cannot_be_changed_after_creation; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-184 - Administrator can open every report with empty business data

Module: Reports | BRD section(s): 15, 18 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed; no operational business values.

Execution steps: Set up/use fixture helpers: seed. Execute requests in method order: GET reports.show.

Expected: Administrator can open every report with empty business data. Assert/check operations: assertOk.

Actual: Passed; 19 assertions; 2.801899 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/ReportsTest.php:16; Tests\Feature\ReportsTest::test_administrator_can_open_every_report_with_empty_business_data; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-185 - Administrator can export reports as xlsx pdf and csv

Module: Reports | BRD section(s): 15, 18 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed; no operational business values.

Execution steps: Set up/use fixture helpers: seed. Execute requests in method order: GET reports.export; GET reports.export; GET reports.export.

Expected: Administrator can export reports as xlsx pdf and csv. Assert/check operations: assertOk, assertHeader.

Actual: Passed; 5 assertions; 0.913566 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/ReportsTest.php:33; Tests\Feature\ReportsTest::test_administrator_can_export_reports_as_xlsx_pdf_and_csv; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-186 - Xlsx export is a valid excel workbook with inline text cells

Module: Reports | BRD section(s): 15, 18 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Invoke services: XlsxReportExporter::create.

Expected: Xlsx export is a valid excel workbook with inline text cells. Assert/check operations: assertTrue, assertNotFalse, assertIsString, assertStringContainsString.

Actual: Passed; 5 assertions; 0.049034 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/ReportsTest.php:47; Tests\Feature\ReportsTest::test_xlsx_export_is_a_valid_excel_workbook_with_inline_text_cells; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-187 - Cashier cannot view financial reports or export reports

Module: Reports | BRD section(s): 15, 18 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed; no operational business values.

Execution steps: Set up/use fixture helpers: seed. Execute requests in method order: GET reports.show; GET reports.export.

Expected: Cashier cannot view financial reports or export reports. Assert/check operations: assertForbidden.

Actual: Passed; 2 assertions; 0.861458 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/ReportsTest.php:65; Tests\Feature\ReportsTest::test_cashier_cannot_view_financial_reports_or_export_reports; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-188 - Retention preserves manual and latest recoverable copy and keeps history

Module: Backups | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and backup; no operational business values.

Execution steps: Set up/use fixture helpers: backup. Invoke services: BackupService::cleanupVerifiedBackups, BackupService::verify.

Expected: Retention preserves manual and latest recoverable copy and keeps history. Assert/check operations: assertTrue, assertFalse, assertNotNull, assertDatabaseCount, assertDatabaseHas.

Actual: Passed; 6 assertions; 0.072562 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Step11BackupTest.php:54; Tests\Feature\Step11BackupTest::test_retention_preserves_manual_and_latest_recoverable_copy_and_keeps_history; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-189 - Changed missing or path traversal backups fail verification

Module: Backups | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and backup; no operational business values.

Execution steps: Set up/use fixture helpers: backup. Invoke services: BackupService::verify.

Expected: Changed missing or path traversal backups fail verification. Assert/check operations: assertTrue, assertFalse, assertArrayNotHasKey.

Actual: Passed; 4 assertions; 0.043910 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Step11BackupTest.php:68; Tests\Feature\Step11BackupTest::test_changed_missing_or_path_traversal_backups_fail_verification; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-190 - Private destination validation rejects all web roots

Module: Backups | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method; no operational business values.

Execution steps: Execute the source method at the cited file/line; it invokes its unit/model/database operation and any declared exception checks.

Expected: Private destination validation rejects all web roots. Assert/check operations: assertFalse.

Actual: Passed; 3 assertions; 0.043474 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Step11BackupTest.php:79; Tests\Feature\Step11BackupTest::test_private_destination_validation_rejects_all_web_roots; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-191 - Google drive stays disabled when configuration is blank

Module: Backups | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and backup; no operational business values.

Execution steps: Set up/use fixture helpers: backup. Invoke services: GoogleDriveBackupService::configured, GoogleDriveBackupService::copy.

Expected: Google drive stays disabled when configuration is blank. Assert/check operations: assertFalse, assertSame.

Actual: Passed; 3 assertions; 0.059935 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Step11BackupTest.php:87; Tests\Feature\Step11BackupTest::test_google_drive_stays_disabled_when_configuration_is_blank; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-192 - Google drive adapter checks remote checksum and size with mocked api

Module: Backups | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and backup; no operational business values.

Execution steps: Set up/use fixture helpers: backup. Invoke services: GoogleDriveBackupService::copy.

Expected: Google drive adapter checks remote checksum and size with mocked api. Assert/check operations: assertSame.

Actual: Passed; 2 assertions; 0.126598 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Step11BackupTest.php:98; Tests\Feature\Step11BackupTest::test_google_drive_adapter_checks_remote_checksum_and_size_with_mocked_api; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-193 - Backup configuration is administrator only and does not reveal paths

Module: Backups | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed, backup; no operational business values.

Execution steps: Set up/use fixture helpers: seed, backup. Execute requests in method order: POST backups.configure; GET /admin/backups; POST backups.configure; GET /admin/backups.

Expected: Backup configuration is administrator only and does not reveal paths. Assert/check operations: assertForbidden, assertSessionHasNoErrors, assertOk, assertDontSee, assertSee, assertDatabaseHas.

Actual: Passed; 7 assertions; 0.995732 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Step11BackupTest.php:112; Tests\Feature\Step11BackupTest::test_backup_configuration_is_administrator_only_and_does_not_reveal_paths; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-194 - All grouped sales statements and refund reports and exports work on empty data

Module: Reports | BRD section(s): 15, 18 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator; no operational business values.

Execution steps: Set up/use fixture helpers: administrator. Execute requests in method order: GET reports.show; GET reports.export.

Expected: All grouped sales statements and refund reports and exports work on empty data. Assert/check operations: assertOk.

Actual: Passed; 26 assertions; 2.497628 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Step11ReportingTest.php:28; Tests\Feature\Step11ReportingTest::test_all_grouped_sales_statements_and_refund_reports_and_exports_work_on_empty_data; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-195 - Collections use actual reversal dates instead of rewriting prior days

Module: Reports | BRD section(s): 15, 18 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator, customer, payment; no operational business values.

Execution steps: Set up/use fixture helpers: administrator, customer, payment.

Expected: Collections use actual reversal dates instead of rewriting prior days. Assert/check operations: assertSame.

Actual: Passed; 3 assertions; 0.857626 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Step11ReportingTest.php:37; Tests\Feature\Step11ReportingTest::test_collections_use_actual_reversal_dates_instead_of_rewriting_prior_days; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-196 - Grouped sales count multi job invoices once and ignore drafts

Module: Reports | BRD section(s): 15, 18 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator, customer, invoice; no operational business values.

Execution steps: Set up/use fixture helpers: administrator, customer, invoice. Execute requests in method order: GET reports.show; GET reports.show. Invoke services: ReportDataService::build.

Expected: Grouped sales count multi job invoices once and ignore drafts. Assert/check operations: assertSame, assertEquals, assertStatus, assertOk.

Actual: Passed; 5 assertions; 2.319203 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Step11ReportingTest.php:53; Tests\Feature\Step11ReportingTest::test_grouped_sales_count_multi_job_invoices_once_and_ignore_drafts; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-197 - Dashboard date cohort and inventory aggregates use real records

Module: Reports | BRD section(s): 15, 18 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator, job, location, part, movement; no operational business values.

Execution steps: Set up/use fixture helpers: administrator, job, location, part, movement. Execute requests in method order: GET dashboard. Invoke services: DashboardService::data.

Expected: Dashboard date cohort and inventory aggregates use real records. Assert/check operations: assertEquals, assertSame, assertCount, assertOk.

Actual: Passed; 6 assertions; 1.078230 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Step11ReportingTest.php:69; Tests\Feature\Step11ReportingTest::test_dashboard_date_cohort_and_inventory_aggregates_use_real_records; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-198 - Cashier own collections and technician reports do not grant financial access

Module: Reports | BRD section(s): 15, 18 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed, employee, payment; no operational business values.

Execution steps: Set up/use fixture helpers: seed, employee, payment. Execute requests in method order: GET reports.show; GET reports.show; GET reports.export. Invoke services: ReportDataService::build.

Expected: Cashier own collections and technician reports do not grant financial access. Assert/check operations: assertSame, assertOk, assertForbidden.

Actual: Passed; 4 assertions; 1.038494 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Step11ReportingTest.php:91; Tests\Feature\Step11ReportingTest::test_cashier_own_collections_and_technician_reports_do_not_grant_financial_access; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-199 - Report pdf and workbook preserve real unicode money and dates

Module: Reports | BRD section(s): 15, 18 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator, customer, job, invoice; no operational business values.

Execution steps: Set up/use fixture helpers: administrator, customer, job, invoice. Execute requests in method order: GET reports.export; GET reports.export.

Expected: Report pdf and workbook preserve real unicode money and dates. Assert/check operations: assertOk, assertStringStartsWith, assertTrue, assertStringContainsString, assertNotFalse.

Actual: Passed; 7 assertions; 1.142748 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Step11ReportingTest.php:106; Tests\Feature\Step11ReportingTest::test_report_pdf_and_workbook_preserve_real_unicode_money_and_dates; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-200 - Customer search treats sql payload as data and escapes stored html

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator, customer; no operational business values.

Execution steps: Set up/use fixture helpers: administrator, customer. Execute requests in method order: GET customers.index; GET customers.show.

Expected: Customer search treats sql payload as data and escapes stored html. Assert/check operations: assertOk, assertViewHas, assertDontSee, assertSee, assertDatabaseCount.

Actual: Passed; 6 assertions; 1.194896 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Step12AuditTest.php:26; Tests\Feature\Step12AuditTest::test_customer_search_treats_sql_payload_as_data_and_escapes_stored_html; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-201 - All report queries execute with linked fixture and record local timings

Module: Audit | BRD section(s): 17 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and administrator, customer, job; no operational business values.

Execution steps: Set up/use fixture helpers: administrator, customer, job. Invoke services: ReportDataService::build.

Expected: All report queries execute with linked fixture and record local timings. Assert/check operations: assertNotEmpty, assertCount, assertOk.

Actual: Passed; 36 assertions; 1.541897 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/Step12AuditTest.php:38; Tests\Feature\Step12AuditTest::test_all_report_queries_execute_with_linked_fixture_and_record_local_timings; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-202 - Purchase grn payment reversal and shared stock integrate without duplicate posting

Module: Purchases | BRD section(s): 13 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and actor, location, part, draft; no operational business values.

Execution steps: Set up/use fixture helpers: actor, location, part, draft. Execute requests in method order: GET purchases.show; GET suppliers.show. Invoke services: InventoryService::availableAtDefaultLocation.

Expected: Purchase grn payment reversal and shared stock integrate without duplicate posting. Assert/check operations: assertSame, assertDatabaseCount, assertOk.

Actual: Passed; 15 assertions; 1.320446 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/StockPurchasingWorkflowTest.php:52; Tests\Feature\StockPurchasingWorkflowTest::test_purchase_grn_payment_reversal_and_shared_stock_integrate_without_duplicate_posting; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-203 - Repair request does not deduct stock and repeated issue deducts once

Module: Purchases | BRD section(s): 13 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and actor, location, part, movement, job; no operational business values.

Execution steps: Set up/use fixture helpers: actor, location, part, movement, job. Execute requests in method order: GET stock-issues.index; GET inventory.show. Invoke services: InventoryService::availableAtDefaultLocation.

Expected: Repair request does not deduct stock and repeated issue deducts once. Assert/check operations: assertDatabaseCount, assertSame, assertOk.

Actual: Passed; 8 assertions; 1.274071 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/StockPurchasingWorkflowTest.php:89; Tests\Feature\StockPurchasingWorkflowTest::test_repair_request_does_not_deduct_stock_and_repeated_issue_deducts_once; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-204 - Stock correction damage count and replay preserve history

Module: Purchases | BRD section(s): 13 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and actor, location, part, movement, expectException; no operational business values.

Execution steps: Set up/use fixture helpers: actor, location, part, movement, expectException. Invoke services: InventoryService::availableAtDefaultLocation.

Expected: Stock correction damage count and replay preserve history. Assert/check operations: assertSame, assertDatabaseCount, expectException.

Actual: Passed; 5 assertions; 1.037009 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/StockPurchasingWorkflowTest.php:112; Tests\Feature\StockPurchasingWorkflowTest::test_stock_correction_damage_count_and_replay_preserve_history; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-205 - New pages render and cashier cannot change stock or suppliers

Module: Purchases | BRD section(s): 13 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and actor; no operational business values.

Execution steps: Set up/use fixture helpers: actor. Execute requests in method order: GET inventory.create; POST inventory.store; POST suppliers.store; POST purchases.store.

Expected: New pages render and cashier cannot change stock or suppliers. Assert/check operations: assertOk, assertForbidden.

Actual: Passed; 11 assertions; 2.508195 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/StockPurchasingWorkflowTest.php:133; Tests\Feature\StockPurchasingWorkflowTest::test_new_pages_render_and_cashier_cannot_change_stock_or_suppliers; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-206 - Http master registration validates precision and records opening balance

Module: Purchases | BRD section(s): 13 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and actor, location, part; no operational business values.

Execution steps: Set up/use fixture helpers: actor, location, part. Execute requests in method order: POST inventory.store; POST suppliers.store; POST inventory.store. Invoke services: InventoryService::latestUnitCost.

Expected: Http master registration validates precision and records opening balance. Assert/check operations: assertSessionHasNoErrors, assertRedirect, assertDatabaseHas, assertSame.

Actual: Passed; 9 assertions; 0.882090 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/StockPurchasingWorkflowTest.php:147; Tests\Feature\StockPurchasingWorkflowTest::test_http_master_registration_validates_precision_and_records_opening_balance; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-207 - Complete unused repair issue can be returned once without deleting history

Module: Purchases | BRD section(s): 13 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and actor, location, part, movement, job; no operational business values.

Execution steps: Set up/use fixture helpers: actor, location, part, movement, job. Invoke services: InventoryService::availableAtDefaultLocation.

Expected: Complete unused repair issue can be returned once without deleting history. Assert/check operations: assertSame, assertDatabaseCount.

Actual: Passed; 5 assertions; 0.852804 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/StockPurchasingWorkflowTest.php:163; Tests\Feature\StockPurchasingWorkflowTest::test_complete_unused_repair_issue_can_be_returned_once_without_deleting_history; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-208 - Receiving more than ordered is rejected atomically

Module: Purchases | BRD section(s): 13 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and actor, location, part, draft, fail; no operational business values.

Execution steps: Set up/use fixture helpers: actor, location, part, draft, fail.

Expected: Receiving more than ordered is rejected atomically. Assert/check operations: assertDatabaseCount.

Actual: Passed; 2 assertions; 0.878838 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/StockPurchasingWorkflowTest.php:182; Tests\Feature\StockPurchasingWorkflowTest::test_receiving_more_than_ordered_is_rejected_atomically; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-209 - Storekeeper can receive but cannot pay or reverse supplier money

Module: Purchases | BRD section(s): 13 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and actor, location, part, draft; no operational business values.

Execution steps: Set up/use fixture helpers: actor, location, part, draft. Execute requests in method order: POST purchases.receive; POST purchases.pay. Invoke services: PurchaseWorkflowService::finalize.

Expected: Storekeeper can receive but cannot pay or reverse supplier money. Assert/check operations: assertSessionHasNoErrors, assertForbidden, assertDatabaseCount.

Actual: Passed; 3 assertions; 1.786298 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/StockPurchasingWorkflowTest.php:199; Tests\Feature\StockPurchasingWorkflowTest::test_storekeeper_can_receive_but_cannot_pay_or_reverse_supplier_money; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-210 - Supplier edit and search preserve the opening ledger

Module: Purchases | BRD section(s): 13 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and actor; no operational business values.

Execution steps: Set up/use fixture helpers: actor. Execute requests in method order: POST suppliers.store; PUT suppliers.update; GET suppliers.index; PUT suppliers.update.

Expected: Supplier edit and search preserve the opening ledger. Assert/check operations: assertSessionHasNoErrors, assertOk, assertViewHas, assertSessionHasErrors, assertSame, assertDatabaseCount.

Actual: Passed; 8 assertions; 0.995922 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/StockPurchasingWorkflowTest.php:212; Tests\Feature\StockPurchasingWorkflowTest::test_supplier_edit_and_search_preserve_the_opening_ledger; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-211 - Unquoted labour charge is authorized audited and used for billing

Module: Purchases | BRD section(s): 13 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and actor, job; no operational business values.

Execution steps: Set up/use fixture helpers: actor, job. Execute requests in method order: POST repair-jobs.labour-charge; POST repair-jobs.labour-charge.

Expected: Unquoted labour charge is authorized audited and used for billing. Assert/check operations: assertSessionHasNoErrors, assertSame, assertDatabaseHas, assertForbidden.

Actual: Passed; 5 assertions; 1.770302 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/StockPurchasingWorkflowTest.php:227; Tests\Feature\StockPurchasingWorkflowTest::test_unquoted_labour_charge_is_authorized_audited_and_used_for_billing; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-212 - Real intake approval stock invoice payment and delivery workflow needs no manual job pricing

Module: Purchases | BRD section(s): 13 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and actor, customer, location, part, movement, code; no operational business values.

Execution steps: Set up/use fixture helpers: actor, customer, location, part, movement, code. Invoke services: RepairReceiptService::register, QuotationService::create, QuotationApprovalService::recordDecision, InvoiceService::create, PaymentService::receiveInvoicePayment, InventoryService::availableAtDefaultLocation, ReportDataService::build.

Expected: Real intake approval stock invoice payment and delivery workflow needs no manual job pricing. Assert/check operations: assertSame, assertDatabaseCount, assertNotNull.

Actual: Passed; 9 assertions; 1.060395 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/StockPurchasingWorkflowTest.php:239; Tests\Feature\StockPurchasingWorkflowTest::test_real_intake_approval_stock_invoice_payment_and_delivery_workflow_needs_no_manual_job_pricing; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-213 - Supplier opening balance payment and reversal use existing supplier ledger

Module: Purchases | BRD section(s): 13 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and actor; no operational business values.

Execution steps: Set up/use fixture helpers: actor. Execute requests in method order: POST suppliers.store; GET suppliers.show.

Expected: Supplier opening balance payment and reversal use existing supplier ledger. Assert/check operations: assertSessionHasNoErrors, assertSame, assertDatabaseCount, assertOk.

Actual: Passed; 6 assertions; 1.021785 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/StockPurchasingWorkflowTest.php:271; Tests\Feature\StockPurchasingWorkflowTest::test_supplier_opening_balance_payment_and_reversal_use_existing_supplier_ledger; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-214 - Supplier return reduces stock and only agreed credit reduces payable

Module: Purchases | BRD section(s): 13 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and actor, location, part, draft; no operational business values.

Execution steps: Set up/use fixture helpers: actor, location, part, draft. Execute requests in method order: GET purchases.show. Invoke services: InventoryService::availableAtDefaultLocation, SupplierPaymentService::outstanding.

Expected: Supplier return reduces stock and only agreed credit reduces payable. Assert/check operations: assertSame, assertDatabaseCount, assertDatabaseHas, assertOk.

Actual: Passed; 7 assertions; 0.994939 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/StockPurchasingWorkflowTest.php:288; Tests\Feature\StockPurchasingWorkflowTest::test_supplier_return_reduces_stock_and_only_agreed_credit_reduces_payable; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-215 - Receipt and card pdf routes render local sinhala and record exports

Module: Purchases | BRD section(s): 13 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and actor, customer, code; no operational business values.

Execution steps: Set up/use fixture helpers: actor, customer, code. Invoke services: RepairReceiptService::register.

Expected: Receipt and card pdf routes render local sinhala and record exports. Assert/check operations: assertOk, assertHeader, assertStringStartsWith, assertDatabaseHas.

Actual: Passed; 10 assertions; 3.518169 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/StockPurchasingWorkflowTest.php:310; Tests\Feature\StockPurchasingWorkflowTest::test_receipt_and_card_pdf_routes_render_local_sinhala_and_record_exports; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-216 - Administrator can update safe business settings and values apply on requests

Module: Settings | BRD section(s): 16, 19 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed; no operational business values.

Execution steps: Set up/use fixture helpers: seed. Execute requests in method order: PUT settings.update; GET settings.index.

Expected: Administrator can update safe business settings and values apply on requests. Assert/check operations: assertRedirect, assertSame, assertDatabaseHas, assertOk, assertSee.

Actual: Passed; 6 assertions; 1.133784 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/SystemSupportTest.php:16; Tests\Feature\SystemSupportTest::test_administrator_can_update_safe_business_settings_and_values_apply_on_requests; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-217 - Business settings reject unknown and secret fields

Module: Settings | BRD section(s): 16, 19 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed; no operational business values.

Execution steps: Set up/use fixture helpers: seed. Execute requests in method order: PUT settings.update.

Expected: Business settings reject unknown and secret fields. Assert/check operations: assertRedirect, assertDatabaseMissing.

Actual: Passed; 2 assertions; 0.954926 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/SystemSupportTest.php:38; Tests\Feature\SystemSupportTest::test_business_settings_reject_unknown_and_secret_fields; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

## T13-218 - Administrator can browse redacted audit history

Module: Settings | BRD section(s): 16, 19 | Status: Passed | Execution date: 2026-10-20

Preconditions: Suite bootstrap and test setUp; isolated MySQL fixtures for database scenarios, in-memory objects for pure unit scenarios.

Test data: Synthetic inputs/fixtures declared in the cited method and seed; no operational business values.

Execution steps: Set up/use fixture helpers: seed. Execute requests in method order: GET audit-logs.index.

Expected: Administrator can browse redacted audit history. Assert/check operations: assertOk, assertSee, assertDontSee.

Actual: Passed; 4 assertions; 0.903193 seconds. JUnit records this exact method/data-set result.

Evidence: tests/Feature/SystemSupportTest.php:58; Tests\Feature\SystemSupportTest::test_administrator_can_browse_redacted_audit_history; documentation/evidence/step13-junit.xml

Defect reference: None raised by this fresh execution.

# 23. Manual acceptance cases

## M13-01 - Physical A5/A6/A4/thermal printing

Module: Printing | BRD reference: acceptance/security/printing/backup/report requirements in matrix | Status: Not Run

Objective / expected: verify physical a5/a6/a4/thermal printing with the approved business environment and retain actual results.

Preconditions / test data: approved isolated UAT environment, actual hardware/account/policy where applicable; non-sensitive fixtures.

Steps: obtain prerequisites; exercise the named workflow under authorized role; compare output/history against the stated requirement and capture evidence.

Actual: Actual shop devices unavailable; inspect paper/scale/signatures and both scripts. Execution date: not executed; status reviewed 2026-10-20.

Evidence: Step 13 completion boundaries and screenshot/test manifests. Defect: related G12 configuration/UAT items where applicable; no fabricated pass.

## M13-02 - Physical barcode scanner

Module: POS | BRD reference: acceptance/security/printing/backup/report requirements in matrix | Status: Not Run

Objective / expected: verify physical barcode scanner with the approved business environment and retain actual results.

Preconditions / test data: approved isolated UAT environment, actual hardware/account/policy where applicable; non-sensitive fixtures.

Steps: obtain prerequisites; exercise the named workflow under authorized role; compare output/history against the stated requirement and capture evidence.

Actual: Browser search/scan field tested; hardware keyboard-wedge scan not exercised. Execution date: not executed; status reviewed 2026-10-20.

Evidence: Step 13 completion boundaries and screenshot/test manifests. Defect: related G12 configuration/UAT items where applicable; no fabricated pass.

## M13-03 - Full end-to-end browser business workflow

Module: System | BRD reference: acceptance/security/printing/backup/report requirements in matrix | Status: Not Run

Objective / expected: verify full end-to-end browser business workflow with the approved business environment and retain actual results.

Preconditions / test data: approved isolated UAT environment, actual hardware/account/policy where applicable; non-sensitive fixtures.

Steps: obtain prerequisites; exercise the named workflow under authorized role; compare output/history against the stated requirement and capture evidence.

Actual: Screenshots verify page rendering; backend scenarios verify posting. No claim of a full browser transaction/UAT test. Execution date: not executed; status reviewed 2026-10-20.

Evidence: Step 13 completion boundaries and screenshot/test manifests. Defect: related G12 configuration/UAT items where applicable; no fabricated pass.

## M13-04 - Wall-clock session expiration

Module: Authentication | BRD reference: acceptance/security/printing/backup/report requirements in matrix | Status: Not Run

Objective / expected: verify wall-clock session expiration with the approved business environment and retain actual results.

Preconditions / test data: approved isolated UAT environment, actual hardware/account/policy where applicable; non-sensitive fixtures.

Steps: obtain prerequisites; exercise the named workflow under authorized role; compare output/history against the stated requirement and capture evidence.

Actual: Configured 120 minutes inspected; waiting through real expiry was not performed. Execution date: not executed; status reviewed 2026-10-20.

Evidence: Step 13 completion boundaries and screenshot/test manifests. Defect: related G12 configuration/UAT items where applicable; no fabricated pass.

## M13-05 - Host unattended 02:00 scheduler execution

Module: Backups | BRD reference: acceptance/security/printing/backup/report requirements in matrix | Status: Not Run

Objective / expected: verify host unattended 02:00 scheduler execution with the approved business environment and retain actual results.

Preconditions / test data: approved isolated UAT environment, actual hardware/account/policy where applicable; non-sensitive fixtures.

Steps: obtain prerequisites; exercise the named workflow under authorized role; compare output/history against the stated requirement and capture evidence.

Actual: No host scheduler activated by this task. Execution date: not executed; status reviewed 2026-10-20.

Evidence: Step 13 completion boundaries and screenshot/test manifests. Defect: related G12 configuration/UAT items where applicable; no fabricated pass.

## M13-06 - Realistic-volume capacity

Module: Performance | BRD reference: acceptance/security/printing/backup/report requirements in matrix | Status: Not Run

Objective / expected: verify realistic-volume capacity with the approved business environment and retain actual results.

Preconditions / test data: approved isolated UAT environment, actual hardware/account/policy where applicable; non-sensitive fixtures.

Steps: obtain prerequisites; exercise the named workflow under authorized role; compare output/history against the stated requirement and capture evidence.

Actual: Local fixture observations are not a load/capacity test. Execution date: not executed; status reviewed 2026-10-20.

Evidence: Step 13 completion boundaries and screenshot/test manifests. Defect: related G12 configuration/UAT items where applicable; no fabricated pass.

## M13-07 - All hostile uploads and every XSS sink

Module: Security | BRD reference: acceptance/security/printing/backup/report requirements in matrix | Status: Not Run

Objective / expected: verify all hostile uploads and every xss sink with the approved business environment and retain actual results.

Preconditions / test data: approved isolated UAT environment, actual hardware/account/policy where applicable; non-sensitive fixtures.

Steps: obtain prerequisites; exercise the named workflow under authorized role; compare output/history against the stated requirement and capture evidence.

Actual: Targeted probes only; no exhaustive penetration test. Execution date: not executed; status reviewed 2026-10-20.

Evidence: Step 13 completion boundaries and screenshot/test manifests. Defect: related G12 configuration/UAT items where applicable; no fabricated pass.

## M13-08 - Printer/scanner business-user sign-off

Module: UAT | BRD reference: acceptance/security/printing/backup/report requirements in matrix | Status: Not Run

Objective / expected: verify printer/scanner business-user sign-off with the approved business environment and retain actual results.

Preconditions / test data: approved isolated UAT environment, actual hardware/account/policy where applicable; non-sensitive fixtures.

Steps: obtain prerequisites; exercise the named workflow under authorized role; compare output/history against the stated requirement and capture evidence.

Actual: Physical test and business tester/date/signature required. Execution date: not executed; status reviewed 2026-10-20.

Evidence: Step 13 completion boundaries and screenshot/test manifests. Defect: related G12 configuration/UAT items where applicable; no fabricated pass.

## M13-09 - Live operational backup destination

Module: Backups | BRD reference: acceptance/security/printing/backup/report requirements in matrix | Status: Blocked

Objective / expected: verify live operational backup destination with the approved business environment and retain actual results.

Preconditions / test data: approved isolated UAT environment, actual hardware/account/policy where applicable; non-sensitive fixtures.

Steps: obtain prerequisites; exercise the named workflow under authorized role; compare output/history against the stated requirement and capture evidence.

Actual: Approved private operational folder has not been supplied. Execution date: not executed; status reviewed 2026-10-20.

Evidence: Step 13 completion boundaries and screenshot/test manifests. Defect: related G12 configuration/UAT items where applicable; no fabricated pass.

## M13-10 - Real Google Drive upload

Module: Backups | BRD reference: acceptance/security/printing/backup/report requirements in matrix | Status: Blocked

Objective / expected: verify real google drive upload with the approved business environment and retain actual results.

Preconditions / test data: approved isolated UAT environment, actual hardware/account/policy where applicable; non-sensitive fixtures.

Steps: obtain prerequisites; exercise the named workflow under authorized role; compare output/history against the stated requirement and capture evidence.

Actual: Folder/account/OAuth intentionally blank at owner request. Execution date: not executed; status reviewed 2026-10-20.

Evidence: Step 13 completion boundaries and screenshot/test manifests. Defect: related G12 configuration/UAT items where applicable; no fabricated pass.

## M13-11 - Purchase-wide freight/discount policy acceptance

Module: Finance | BRD reference: acceptance/security/printing/backup/report requirements in matrix | Status: Blocked

Objective / expected: verify purchase-wide freight/discount policy acceptance with the approved business environment and retain actual results.

Preconditions / test data: approved isolated UAT environment, actual hardware/account/policy where applicable; non-sensitive fixtures.

Steps: obtain prerequisites; exercise the named workflow under authorized role; compare output/history against the stated requirement and capture evidence.

Actual: Owner chose latest purchase cost; allocation of purchase-wide adjustments remains undecided. Execution date: not executed; status reviewed 2026-10-20.

Evidence: Step 13 completion boundaries and screenshot/test manifests. Defect: related G12 configuration/UAT items where applicable; no fabricated pass.

### M13-12 - Final business-owner UAT approval

Module: UAT | BRD reference: acceptance/security/printing/backup/report requirements in matrix | Status: Blocked

Objective / expected: verify final business-owner uat approval with the approved business environment and retain actual results.

Preconditions / test data: approved isolated UAT environment, actual hardware/account/policy where applicable; non-sensitive fixtures.

Steps: obtain prerequisites; exercise the named workflow under authorized role; compare output/history against the stated requirement and capture evidence.

Actual: No owner signature/acceptance supplied; automation cannot provide it. Execution date: not executed; status reviewed 2026-10-20.

Evidence: Step 13 completion boundaries and screenshot/test manifests. Defect: related G12 configuration/UAT items where applicable; no fabricated pass.

# 24. Requirement-by-requirement appendix

All maintained BRD IDs are retained below. Status is the matrix requirement status, not one of the execution statuses used in test cases. Field-only/manual gaps remain visible. The original full 12-column Markdown matrix is preserved as companion evidence.

## BRD section 1

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-01-01 / Mandatory | Integrated repair and retail system / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Core operational workflows and selected cross-module scenarios now pass. Full owner/hardware/configuration acceptance remains pending. |
| BRD-01-02 / Mandatory | Permanent central customer and transaction records / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Core operational workflows and selected cross-module scenarios now pass. Full owner/hardware/configuration acceptance remains pending. |

## BRD section 2

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-02-01 / Mandatory | All twelve business objectives / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; See individual objectives in sections 4–19; no blanket completion claim. |

## BRD section 3

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-03-01 / Mandatory | Fourteen BRD modules / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Core operational workflows and selected cross-module scenarios now pass. Full owner/hardware/configuration acceptance remains pending. |

## BRD section 4

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-04-01 / Mandatory | Individual registration / Customers | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-04-02 / Mandatory | Company registration / Customers | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-04-03 / Mandatory | Automatic unique customer ID / Customers | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-04-04 / Mandatory | Required name address and mobile validation / Customers | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-04-05 / Mandatory | Search by name / Customers | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-04-06 / Mandatory | Search by mobile / Customers | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-04-07 / Mandatory | Search by customer code / Customers | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-04-08 / Mandatory | Customer editing / Customers | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-04-09 / Mandatory | Activation and deactivation / Customers | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-04-10 / Mandatory | Customer history links / Customers | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-04-11 / Mandatory | Customer statement and credit ledger / Customers | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-04-12 / Mandatory | Contact person / Customers | Implemented but Not Tested; Not verified end to end | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; Fields exist; each field has not received a separate manual UAT pass. |
| BRD-04-13 / Mandatory | Alternative telephone / Customers | Implemented but Not Tested; Not verified end to end | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; Fields exist; each field has not received a separate manual UAT pass. |
| BRD-04-14 / Mandatory | Email / Customers | Implemented but Not Tested; Not verified end to end | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; Fields exist; each field has not received a separate manual UAT pass. |
| BRD-04-15 / Mandatory | Registration date / Customers | Implemented but Not Tested; Not verified end to end | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; Fields exist; each field has not received a separate manual UAT pass. |
| BRD-04-16 / Mandatory | Notes / Customers | Implemented but Not Tested; Not verified end to end | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; Fields exist; each field has not received a separate manual UAT pass. |
| BRD-04-17 / Mandatory | Authorized opening balance posted once / Customers | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/CustomerManagementTest.php; tests/Feature/PaymentManagementTest.php; |

## BRD section 5

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-05-01 / Mandatory | Atomic receipt and job creation / RepairReceipts | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-05-02 / Mandatory | Automatic Job ID / RepairReceipts | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-05-03 / Mandatory | Unique manual job number / RepairReceipts | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-05-04 / Mandatory | Appliance and customer selection / RepairReceipts | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-05-05 / Mandatory | Required brand or model / RepairReceipts | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-05-06 / Mandatory | Advance payment linkage / RepairReceipts | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-05-07 / Mandatory | Registered receiver validation / RepairReceipts | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-05-08 / Mandatory | Labour cost / RepairReceipts | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-05-09 / Mandatory | A5 receipt / RepairReceipts | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-05-10 / Mandatory | Duplicate reprint and history / RepairReceipts | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-05-11 / Mandatory | Job type excluded from receipt output / RepairReceipts | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-05-12 / Mandatory | Rollback on failed advance / RepairReceipts | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-05-13 / Mandatory | Received timestamp / RepairReceipts | Implemented but Not Tested; Not verified end to end | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; Present in intake/snapshot or printed layout; physical acceptance pending. |
| BRD-05-14 / Mandatory | Reported fault / RepairReceipts | Implemented but Not Tested; Not verified end to end | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; Present in intake/snapshot or printed layout; physical acceptance pending. |
| BRD-05-15 / Mandatory | Accessories / RepairReceipts | Implemented but Not Tested; Not verified end to end | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; Present in intake/snapshot or printed layout; physical acceptance pending. |
| BRD-05-16 / Mandatory | Physical condition / RepairReceipts | Implemented but Not Tested; Not verified end to end | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; Present in intake/snapshot or printed layout; physical acceptance pending. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-05-17 / Mandatory | Visible damage / RepairReceipts | Implemented but Not Tested; Not verified end to end | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; Present in intake/snapshot or printed layout; physical acceptance pending. |
| BRD-05-18 / Mandatory | Signature area / RepairReceipts | Implemented but Not Tested; Not verified end to end | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; Present in intake/snapshot or printed layout; physical acceptance pending. |
| BRD-05-19 / Mandatory | Special notes / RepairReceipts | Implemented but Not Tested; Not verified end to end | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/PaymentManagementTest.php; Present in intake/snapshot or printed layout; physical acceptance pending. |

## BRD section 6

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-06-01 / Mandatory | Immutable job identifiers / RepairJobs | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/Database/BusinessConstraintsTest.php; |
| BRD-06-02 / Mandatory | Technician assignment / RepairJobs | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/Database/BusinessConstraintsTest.php; |
| BRD-06-03 / Mandatory | Assigned technician authorization / RepairJobs | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/Database/BusinessConstraintsTest.php; |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-06-04 / Mandatory | Inspection records / RepairJobs | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/Database/BusinessConstraintsTest.php; |
| BRD-06-05 / Mandatory | Repair notes / RepairJobs | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/Database/BusinessConstraintsTest.php; |
| BRD-06-06 / Mandatory | All seventeen status values / RepairJobs | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/Database/BusinessConstraintsTest.php; |
| BRD-06-07 / Mandatory | Append-only status history / RepairJobs | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/Database/BusinessConstraintsTest.php; |
| BRD-06-08 / Mandatory | Changed-by actor and timestamp / RepairJobs | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/Database/BusinessConstraintsTest.php; |
| BRD-06-09 / Mandatory | A6 two-page writable job card / RepairJobs | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/Database/BusinessConstraintsTest.php; |
| BRD-06-10 / Mandatory | Parts used with quantity and cost / RepairJobs | Implemented and Verified; Passed complete workflow and native race checks | tests/Feature/StockPurchasingWorkflowTest.php; scripts/verify-pending-concurrency.php; StockPurchasingWorkflowTest and verify-pending-concurrency.php supersede the initial missing-workflow finding. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-06-11 / Mandatory | Completion and delivery workflow / RepairJobs | Implemented and Verified; Passed complete workflow and native race checks | tests/Feature/StockPurchasingWorkflowTest.php; scripts/verify-pending-concurrency.php; StockPurchasingWorkflowTest and verify-pending-concurrency.php supersede the initial missing-workflow finding. |

## BRD section 7

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-07-01 / Mandatory | Create quotation / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-02 / Mandatory | Edit draft / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-03 / Mandatory | Unique quotation number / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-04 / Mandatory | Job and customer link / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-05 / Mandatory | Part quantity and unit pricing / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-06 / Mandatory | Exact labour and total / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-07 / Mandatory | Validity validation / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-08 / Mandatory | Issue snapshot / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-09 / Mandatory | Customer approval / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-07-10 / Mandatory | Customer rejection / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-11 / Mandatory | Approval history / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-12 / Mandatory | Revision and supersession / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-13 / Mandatory | Approval-gated repair start / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-14 / Mandatory | No stock deduction on approval / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-15 / Mandatory | No payment on approval / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-16 / Mandatory | A4 PDF / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-17 / Mandatory | A5 PDF / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-18 / Mandatory | Approved values reused by invoice / Quotations | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; |
| BRD-07-19 / Mandatory | Owner acceptance of terms and signature layout / Quotations | Implemented but Not Tested; Not verified end to end | tests/Feature/QuotationManagementTest.php; tests/Feature/InvoiceManagementTest.php; Physical layout approval not supplied. |

## BRD section 8

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-08-01 / Mandatory | Search automatic and manual job number / RepairJobs | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/Database/BusinessConstraintsTest.php; |
| BRD-08-02 / Mandatory | Search customer and mobile / RepairJobs | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/Database/BusinessConstraintsTest.php; |
| BRD-08-03 / Mandatory | Search item / RepairJobs | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/Database/BusinessConstraintsTest.php; |
| BRD-08-04 / Mandatory | Status filters / RepairJobs | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/Database/BusinessConstraintsTest.php; |
| BRD-08-05 / Mandatory | Paginated job details / RepairJobs | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/Database/BusinessConstraintsTest.php; |
| BRD-08-06 / Mandatory | Job totals advances and outstanding visibility / RepairJobs | Partially Implemented; Not verified end to end | tests/Feature/RepairReceiptAndJobManagementTest.php; tests/Feature/Database/BusinessConstraintsTest.php; Invoice balances tested; complete job-summary presentation not separately verified. |

## BRD section 9

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-09-01 / Mandatory | Automatic invoice number / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-09-02 / Mandatory | Single job invoice / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-09-03 / Mandatory | Multiple jobs per invoice / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-09-04 / Mandatory | Same customer enforced / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-09-05 / Mandatory | Prevent duplicate job billing / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-09-06 / Mandatory | Job-wise exact charges / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-09-07 / Mandatory | Approved quote snapshot / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-09-08 / Mandatory | Invoice total / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-09-09 / Mandatory | Advance and allocation balance / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-09-10 / Mandatory | Cash and credit terms / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-09-11 / Mandatory | Cashier identity / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-09-12 / Mandatory | A5 print / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-09-13 / Mandatory | Duplicate print / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-09-14 / Mandatory | PDF export / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-09-15 / Mandatory | Cancellation history / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-09-16 / Mandatory | Issued snapshot immutable / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-09-17 / Mandatory | Failed line insertion rolls back transaction / Invoices | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; |
| BRD-09-18 / Mandatory | Collector / Invoices | Implemented but Not Tested; Not verified end to end | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; Fields persist; individual business acceptance pending. |
| BRD-09-19 / Mandatory | Vehicle number / Invoices | Implemented but Not Tested; Not verified end to end | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; Fields persist; individual business acceptance pending. |
| BRD-09-20 / Mandatory | Delivery date / Invoices | Implemented but Not Tested; Not verified end to end | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; Fields persist; individual business acceptance pending. |
| BRD-09-21 / Mandatory | Delivery information / Invoices | Implemented but Not Tested; Not verified end to end | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; Fields persist; individual business acceptance pending. |
| BRD-09-22 / Mandatory | Signature spaces / Invoices | Implemented but Not Tested; Not verified end to end | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; Fields persist; individual business acceptance pending. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-09-23 / Mandatory | Special notes / Invoices | Implemented but Not Tested; Not verified end to end | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; Fields persist; individual business acceptance pending. |
| BRD-09-24 / Mandatory | Current payment during invoice workflow / Invoices | Partially Implemented; Not verified end to end | tests/Feature/InvoiceManagementTest.php; tests/Feature/PaymentManagementTest.php; Invoice issue and payment capture are separate authorized operations; cancellation retains job reservation. |

## BRD section 10

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-10-01 / Mandatory | Cash payment / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-10-02 / Mandatory | Bank transfer / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-10-03 / Mandatory | Cheque payment / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-10-04 / Mandatory | Customer advance / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-10-05 / Mandatory | Job advance / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-10-06 / Mandatory | Partial payment / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-10-07 / Mandatory | Repeated payments / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-10-08 / Mandatory | Idempotent payment / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-10-09 / Mandatory | Cross-customer allocation rejection / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-10-10 / Mandatory | Derived outstanding / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-10-11 / Mandatory | Net reversals / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-10-12 / Mandatory | Authorized refund / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-10-13 / Mandatory | Reversal audit / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-10-14 / Mandatory | Decimal precision / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-10-15 / Mandatory | No double-counted advance / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-10-16 / Mandatory | Payment receipt / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |

## BRD section 11

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-11-01 / Mandatory | Credit invoice list / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-11-02 / Mandatory | Credit collection permissions / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-11-03 / Mandatory | Partial credit collections / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-11-04 / Mandatory | Outstanding balance / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-11-05 / Mandatory | Customer account statement / Payments | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PaymentManagementTest.php; |
| BRD-11-06 / Mandatory | Owner acceptance of overdue credit policy / Payments | Implemented but Not Tested; Not verified end to end | tests/Feature/PaymentManagementTest.php; Due-date filter exists; no invented overdue escalation policy. |

## BRD section 12

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-12-01 / Mandatory | One shared repair and POS inventory / Inventory | Implemented and Verified; Passed native shared-stock tests | tests/Feature/Database/BusinessConstraintsTest.php; tests/Feature/POSManagementTest.php; Same ledger and product lock are used by POS, GRNs, adjustments, counts and repair issues. |
| BRD-12-02 / Mandatory | Spare part master schema / Inventory | Partially Implemented; Not verified end to end | tests/Feature/Database/BusinessConstraintsTest.php; tests/Feature/POSManagementTest.php; Operational stock workflows are implemented. Remaining limitations concern individual manual acceptance and special costing/return cases, not absence of the module. |
| BRD-12-03 / Mandatory | Unique SKU and barcode / Inventory | Partially Implemented; Not verified end to end | tests/Feature/Database/BusinessConstraintsTest.php; tests/Feature/POSManagementTest.php; Operational stock workflows are implemented. Remaining limitations concern individual manual acceptance and special costing/return cases, not absence of the module. |
| BRD-12-04 / Mandatory | Opening and current stock / Inventory | Implemented and Verified; Passed native shared-stock tests | tests/Feature/Database/BusinessConstraintsTest.php; tests/Feature/POSManagementTest.php; Same ledger and product lock are used by POS, GRNs, adjustments, counts and repair issues. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-12-05 / Mandatory | Stock movement history / Inventory | Implemented and Verified; Passed native shared-stock tests | tests/Feature/Database/BusinessConstraintsTest.php; tests/Feature/POSManagementTest.php; Same ledger and product lock are used by POS, GRNs, adjustments, counts and repair issues. |
| BRD-12-06 / Mandatory | Low stock and reorder alerts / Inventory | Partially Implemented; Not verified end to end | tests/Feature/Database/BusinessConstraintsTest.php; tests/Feature/POSManagementTest.php; Operational stock workflows are implemented. Remaining limitations concern individual manual acceptance and special costing/return cases, not absence of the module. |
| BRD-12-07 / Mandatory | Stock valuation / Inventory | Implemented and Verified; Latest purchase cost and old-return regression passed | tests/Feature/LatestPurchaseCostTest.php; Per-location latest posted purchase cost, with explicit known opening-cost fallback. Unknown costs are not valued at zero. |
| BRD-12-08 / Mandatory | Spare part registration UI / Inventory | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-12-09 / Mandatory | Stock receiving workflow / Inventory | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-12-10 / Mandatory | Approved repair-part issue / Inventory | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-12-11 / Mandatory | Supplier stock return workflow / Inventory | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-12-12 / Mandatory | Controlled stock adjustments / Inventory | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-12-13 / Mandatory | Damage posting workflow / Inventory | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-12-14 / Mandatory | Physical stock count posting / Inventory | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-12-15 / Mandatory | Repair issue cancellation and undo / Inventory | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-12-16 / Mandatory | POS deduction once / Inventory | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/Database/BusinessConstraintsTest.php; tests/Feature/POSManagementTest.php; |
| BRD-12-17 / Mandatory | POS resalable return / Inventory | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/Database/BusinessConstraintsTest.php; tests/Feature/POSManagementTest.php; |
| BRD-12-18 / Mandatory | No quotation stock deduction / Inventory | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/Database/BusinessConstraintsTest.php; tests/Feature/POSManagementTest.php; |
| BRD-12-19 / Mandatory | No print stock deduction / Inventory | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/Database/BusinessConstraintsTest.php; tests/Feature/POSManagementTest.php; |
| BRD-12-20 / Mandatory | POS overselling prevention / Inventory | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/Database/BusinessConstraintsTest.php; tests/Feature/POSManagementTest.php; |
| BRD-12-21 / Optional | Stock transfer between locations / Inventory | Not Applicable; Not executed / unavailable | tests/Feature/Database/BusinessConstraintsTest.php; tests/Feature/POSManagementTest.php; Optional in BRD; not implemented or required in this verification scope. |

# BRD section 13

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-13-01 / Mandatory | Supplier registration / Suppliers | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-13-02 / Mandatory | Supplier editing / Suppliers | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-13-03 / Mandatory | Supplier search / Suppliers | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-13-04 / Mandatory | Supplier credit terms workflow / Suppliers | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-13-05 / Mandatory | Supplier payment entry / Suppliers | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-13-06 / Mandatory | Partial supplier payment / Suppliers | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-13-07 / Mandatory | Supplier payment reversal / Suppliers | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-13-08 / Mandatory | Supplier purchase history workflow / Suppliers | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-13-09 / Mandatory | Purchase draft entry / Purchases | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-13-10 / Mandatory | Purchase item entry / Purchases | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-13-11 / Mandatory | Discount and other-cost allocation / Purchases | Partially Implemented; Not executed / unavailable | tests/Feature/Database/BusinessRelationshipsTest.php; Purchase total includes discount/other costs; unit-cost allocation awaits the owner decision. |
| BRD-13-12 / Mandatory | GRN creation / Purchases | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-13-13 / Mandatory | Partial receiving / Purchases | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-13-14 / Mandatory | GRN stock increase once / Purchases | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-13-15 / Mandatory | Supplier purchase credit settlement / Purchases | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up implementation preserves existing ledgers and history. Repair returns currently undo a complete unused issue; duplicate-part purchase lines require separate return reconciliation. |
| BRD-13-16 / Mandatory | Supplier statement / Reports | Partially Implemented; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Source workflows now exist and core reports execute. Each populated filter/export permutation and unresolved cost allocation are not certified by the general suite. |
| BRD-13-17 / Mandatory | Supplier outstanding report / Reports | Partially Implemented; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Source workflows now exist and core reports execute. Each populated filter/export permutation and unresolved cost allocation are not certified by the general suite. |
| BRD-13-18 / Mandatory | Purchase report / Reports | Partially Implemented; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Source workflows now exist and core reports execute. Each populated filter/export permutation and unresolved cost allocation are not certified by the general suite. |
| BRD-13-19 / Mandatory | Supplier and purchase management access policy agreement / Employees | Blocked; Not executed / unavailable | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; Section 13 says management only; section 14 also grants storekeeper purchases. Owner decision remains open. |

## BRD section 14

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-14-01 / Mandatory | Employee registration / Employees | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; |
| BRD-14-02 / Mandatory | Unique employee code / Employees | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; |
| BRD-14-03 / Mandatory | Hashed login account / Employees | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; |
| BRD-14-04 / Mandatory | Case-insensitive username uniqueness / Employees | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; |
| BRD-14-05 / Mandatory | Employee active state / Employees | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; |
| BRD-14-06 / Mandatory | Administrator role / Employees | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; |
| BRD-14-07 / Mandatory | Receptionist role / Employees | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; |
| BRD-14-08 / Mandatory | Technician role / Employees | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; |
| BRD-14-09 / Mandatory | Cashier role / Employees | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; |
| BRD-14-10 / Mandatory | Storekeeper role / Employees | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; |
| BRD-14-11 / Mandatory | Manager role / Employees | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; |
| BRD-14-12 / Mandatory | Per-user permission denial / Employees | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-14-13 / Mandatory | Backend route authorization / Employees | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; |
| BRD-14-14 / Mandatory | Assigned-job record authorization / Employees | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; |
| BRD-14-15 / Mandatory | Last administrator protection / Employees | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; |
| BRD-14-16 / Mandatory | Employee address phone title department and join date / Employees | Implemented but Not Tested; Not verified end to end | tests/Feature/EmployeeManagementTest.php; tests/Feature/Auth/AuthenticationTest.php; Model/form fields exist; separate field-by-field UAT not performed. |
| BRD-14-17 / Mandatory | Storekeeper purchase and stock operations / Employees | Implemented and Verified; Passed complete workflow and native race checks | tests/Feature/StockPurchasingWorkflowTest.php; scripts/verify-pending-concurrency.php; StockPurchasingWorkflowTest and verify-pending-concurrency.php supersede the initial missing-workflow finding. |

## BRD section 15

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-15-01 / Mandatory | Daily jobs report / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-02 / Mandatory | Pending job status filter / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-15-03 / Mandatory | Under repair status filter / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-04 / Mandatory | Completed job status filter / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-05 / Mandatory | Delivered job status filter / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-06 / Mandatory | Technician jobs report / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-07 / Mandatory | Customer filter / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-15-08 / Mandatory | Date range filter / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-09 / Mandatory | Daily sales / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-10 / Mandatory | Monthly sales / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-11 / Mandatory | Yearly sales / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-12 / Mandatory | Invoice sales / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-15-13 / Mandatory | Repair income report / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-14 / Mandatory | POS sales / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-15 / Mandatory | Cash and credit report / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-16 / Mandatory | Cashier sales / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-17 / Mandatory | Payment collections / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-15-18 / Mandatory | Advance report / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-19 / Mandatory | Customer outstanding / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-20 / Mandatory | Overdue credit / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-21 / Mandatory | Discounts / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-22 / Mandatory | Refunds on reversal date / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-15-23 / Mandatory | Report PDF / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-24 / Mandatory | Excel workbook export / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-25 / Mandatory | CSV export / Reports | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Query execution and report suite assertions; not every populated report combination has a separate manual pass. |
| BRD-15-26 / Mandatory | Waiting for parts filter / Reports | Implemented but Not Tested; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Implemented report/filter code; these individual populated combinations were not separately asserted in this audit. |
| BRD-15-27 / Mandatory | Ready for collection filter / Reports | Implemented but Not Tested; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Implemented report/filter code; these individual populated combinations were not separately asserted in this audit. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-15-28 / Mandatory | Cancelled jobs filter / Reports | Implemented but Not Tested; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Implemented report/filter code; these individual populated combinations were not separately asserted in this audit. |
| BRD-15-29 / Mandatory | Unable to repair filter / Reports | Implemented but Not Tested; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Implemented report/filter code; these individual populated combinations were not separately asserted in this audit. |
| BRD-15-30 / Mandatory | Completion time report / Reports | Implemented but Not Tested; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Implemented report/filter code; these individual populated combinations were not separately asserted in this audit. |
| BRD-15-31 / Mandatory | Daily cash summary / Reports | Implemented but Not Tested; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Implemented report/filter code; these individual populated combinations were not separately asserted in this audit. |
| BRD-15-32 / Mandatory | Daily cash collection / Reports | Implemented but Not Tested; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Implemented report/filter code; these individual populated combinations were not separately asserted in this audit. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-15-33 / Mandatory | Status filter / Reports | Implemented but Not Tested; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Implemented report/filter code; these individual populated combinations were not separately asserted in this audit. |
| BRD-15-34 / Mandatory | Employee filter / Reports | Implemented but Not Tested; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Implemented report/filter code; these individual populated combinations were not separately asserted in this audit. |
| BRD-15-35 / Mandatory | Item and category filter / Reports | Implemented but Not Tested; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Implemented report/filter code; these individual populated combinations were not separately asserted in this audit. |
| BRD-15-36 / Mandatory | Browser report print / Reports | Implemented but Not Tested; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Implemented report/filter code; these individual populated combinations were not separately asserted in this audit. |
| BRD-15-37 / Mandatory | Current stock report / Reports | Partially Implemented; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Source workflows now exist and core reports execute. Each populated filter/export permutation and unresolved cost allocation are not certified by the general suite. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-15-38 / Mandatory | Stock valuation / Reports | Implemented and Verified; Latest purchase cost and old-return regression passed | tests/Feature/LatestPurchaseCostTest.php; Per-location latest posted purchase cost, with explicit known opening-cost fallback. Unknown costs are not valued at zero. |
| BRD-15-39 / Mandatory | Low stock / Reports | Partially Implemented; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Source workflows now exist and core reports execute. Each populated filter/export permutation and unresolved cost allocation are not certified by the general suite. |
| BRD-15-40 / Mandatory | Out of stock / Reports | Partially Implemented; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Source workflows now exist and core reports execute. Each populated filter/export permutation and unresolved cost allocation are not certified by the general suite. |
| BRD-15-41 / Mandatory | Stock movement report / Reports | Partially Implemented; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Source workflows now exist and core reports execute. Each populated filter/export permutation and unresolved cost allocation are not certified by the general suite. |
| BRD-15-42 / Mandatory | Repair parts report / Reports | Partially Implemented; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Source workflows now exist and core reports execute. Each populated filter/export permutation and unresolved cost allocation are not certified by the general suite. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-15-43 / Mandatory | POS parts report / Reports | Partially Implemented; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Source workflows now exist and core reports execute. Each populated filter/export permutation and unresolved cost allocation are not certified by the general suite. |
| BRD-15-44 / Mandatory | Stock adjustment report / Reports | Partially Implemented; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Source workflows now exist and core reports execute. Each populated filter/export permutation and unresolved cost allocation are not certified by the general suite. |
| BRD-15-45 / Mandatory | Purchase history / Reports | Partially Implemented; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Source workflows now exist and core reports execute. Each populated filter/export permutation and unresolved cost allocation are not certified by the general suite. |
| BRD-15-46 / Mandatory | Supplier-wise purchases / Reports | Partially Implemented; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Source workflows now exist and core reports execute. Each populated filter/export permutation and unresolved cost allocation are not certified by the general suite. |
| BRD-15-47 / Mandatory | Supplier outstanding / Reports | Partially Implemented; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Source workflows now exist and core reports execute. Each populated filter/export permutation and unresolved cost allocation are not certified by the general suite. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-15-48 / Mandatory | Purchase summary / Reports | Partially Implemented; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Source workflows now exist and core reports execute. Each populated filter/export permutation and unresolved cost allocation are not certified by the general suite. |
| BRD-15-49 / Mandatory | Supplier statement / Reports | Partially Implemented; Not verified end to end | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Source workflows now exist and core reports execute. Each populated filter/export permutation and unresolved cost allocation are not certified by the general suite. |
| BRD-15-50 / Mandatory | Profit estimate / Reports | Partially Implemented; Recorded-cost repair/POS reconciliation passed | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; Source workflows now exist and core reports execute. Each populated filter/export permutation and unresolved cost allocation are not certified by the general suite. |
| BRD-15-51 / Mandatory | Expense entry and expense report business acceptance / Reports | Not Applicable; Not executed / unavailable | tests/Feature/ReportsTest.php; tests/Feature/Step11ReportingTest.php; tests/Feature/Step12AuditTest.php; BRD conditional on an implemented expense workflow; schema/report query alone does not establish that workflow. |

## BRD section 16

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-16-01 / Mandatory | Primary keys / System | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Read-only operational metadata plus disposable constraint tests. |
| BRD-16-02 / Mandatory | Foreign keys / System | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Read-only operational metadata plus disposable constraint tests. |
| BRD-16-03 / Mandatory | Unique invoice-job reservation / System | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Read-only operational metadata plus disposable constraint tests. |
| BRD-16-04 / Mandatory | Same-customer associations / System | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Read-only operational metadata plus disposable constraint tests. |
| BRD-16-05 / Mandatory | Append-only transaction history / System | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Read-only operational metadata plus disposable constraint tests. |
| BRD-16-06 / Mandatory | Transactional receipt and invoice rollback / System | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Read-only operational metadata plus disposable constraint tests. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-16-07 / Mandatory | Decimal financial storage / System | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Read-only operational metadata plus disposable constraint tests. |
| BRD-16-08 / Mandatory | InnoDB / System | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Read-only operational metadata plus disposable constraint tests. |
| BRD-16-09 / Mandatory | utf8mb4 / System | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Read-only operational metadata plus disposable constraint tests. |
| BRD-16-10 / Mandatory | Complete integrated transaction lifecycle / System | Implemented and Verified; Passed complete workflow and native race checks | tests/Feature/StockPurchasingWorkflowTest.php; scripts/verify-pending-concurrency.php; StockPurchasingWorkflowTest and verify-pending-concurrency.php supersede the initial missing-workflow finding. |

## BRD section 17

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-17-01 / Mandatory | Manual backup service / BackupRestore | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/Step11BackupTest.php; tests/Feature/BackupManagementTest.php; scripts/verify-step11-recovery.php; Real recovery helper uses disposable database and temporary storage only. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-17-02 / Mandatory | Backup file verification / BackupRestore | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/Step11BackupTest.php; tests/Feature/BackupManagementTest.php; scripts/verify-step11-recovery.php; Real recovery helper uses disposable database and temporary storage only. |
| BRD-17-03 / Mandatory | History / BackupRestore | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/Step11BackupTest.php; tests/Feature/BackupManagementTest.php; scripts/verify-step11-recovery.php; Real recovery helper uses disposable database and temporary storage only. |
| BRD-17-04 / Mandatory | Private-path restriction / BackupRestore | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/Step11BackupTest.php; tests/Feature/BackupManagementTest.php; scripts/verify-step11-recovery.php; Real recovery helper uses disposable database and temporary storage only. |
| BRD-17-05 / Mandatory | Retention logic / BackupRestore | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/Step11BackupTest.php; tests/Feature/BackupManagementTest.php; scripts/verify-step11-recovery.php; Real recovery helper uses disposable database and temporary storage only. |
| BRD-17-06 / Mandatory | Failure handling / BackupRestore | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/Step11BackupTest.php; tests/Feature/BackupManagementTest.php; scripts/verify-step11-recovery.php; Real recovery helper uses disposable database and temporary storage only. |
| BRD-17-07 / Mandatory | Restore confirmation / BackupRestore | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/Step11BackupTest.php; tests/Feature/BackupManagementTest.php; scripts/verify-step11-recovery.php; Real recovery helper uses disposable database and temporary storage only. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-17-08 / Mandatory | Administrator password check / BackupRestore | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/Step11BackupTest.php; tests/Feature/BackupManagementTest.php; scripts/verify-step11-recovery.php; Real recovery helper uses disposable database and temporary storage only. |
| BRD-17-09 / Mandatory | Real disposable restore / BackupRestore | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/Step11BackupTest.php; tests/Feature/BackupManagementTest.php; scripts/verify-step11-recovery.php; Real recovery helper uses disposable database and temporary storage only. |
| BRD-17-10 / Mandatory | Restore audit / BackupRestore | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/Step11BackupTest.php; tests/Feature/BackupManagementTest.php; scripts/verify-step11-recovery.php; Real recovery helper uses disposable database and temporary storage only. |
| BRD-17-11 / Mandatory | Automatic daily backup / BackupRestore | Partially Implemented; Not verified end to end | tests/Feature/Step11BackupTest.php; tests/Feature/BackupManagementTest.php; scripts/verify-step11-recovery.php; Schedule definition 02:00 Asia/Colombo exists. Destination and Drive credentials blank; host scheduler not activated by this task. |
| BRD-17-12 / Mandatory | Backup destination / BackupRestore | Partially Implemented; Not verified end to end | tests/Feature/Step11BackupTest.php; tests/Feature/BackupManagementTest.php; scripts/verify-step11-recovery.php; Schedule definition 02:00 Asia/Colombo exists. Destination and Drive credentials blank; host scheduler not activated by this task. |
| BRD-17-13 / Mandatory | Off-computer backup / BackupRestore | Partially Implemented; Not verified end to end | tests/Feature/Step11BackupTest.php; tests/Feature/BackupManagementTest.php; scripts/verify-step11-recovery.php; Schedule definition 02:00 Asia/Colombo exists. Destination and Drive credentials blank; host scheduler not activated by this task. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-17-14 / Mandatory | Password hashing / System | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; |
| BRD-17-15 / Mandatory | Login rate limit / System | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; |
| BRD-17-16 / Mandatory | CSRF protection / System | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; |
| BRD-17-17 / Mandatory | Active-account middleware / System | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; |
| BRD-17-18 / Mandatory | Audit access control / System | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; |
| BRD-17-19 / Optional | Configurable idle session timeout / System | Implemented but Not Tested; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Optional; session defaults reviewed but no time-elapse acceptance run. |

## BRD section 18

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-18-01 / Mandatory | Windows multiuser application / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Local app and two-process races verified; physical scanner/printer and complete browser/responsive sweep not executed. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-18-02 / Mandatory | Central MySQL / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Local app and two-process races verified; physical scanner/printer and complete browser/responsive sweep not executed. |
| BRD-18-03 / Mandatory | English and Sinhala UI / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Local app and two-process races verified; physical scanner/printer and complete browser/responsive sweep not executed. |
| BRD-18-04 / Mandatory | Scanner support / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Local app and two-process races verified; physical scanner/printer and complete browser/responsive sweep not executed. |
| BRD-18-05 / Mandatory | A4 A5 A6 output / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Local app and two-process races verified; physical scanner/printer and complete browser/responsive sweep not executed. |
| BRD-18-06 / Mandatory | PDF and Excel support / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Local app and two-process races verified; physical scanner/printer and complete browser/responsive sweep not executed. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-18-07 / Mandatory | User-friendly responsive interface / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Local app and two-process races verified; physical scanner/printer and complete browser/responsive sweep not executed. |
| BRD-18-08 / Mandatory | Production hosting and online deployment / System | Not Applicable; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Explicitly excluded from Step 12; no deployment performed. |

## BRD section 19

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-19-01 / Mandatory | Versioned template settings / PrintTemplates | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PrintTemplateManagementTest.php; tests/Feature/InvoiceManagementTest.php; tests/Feature/POSManagementTest.php; |
| BRD-19-02 / Mandatory | Authorized template edits / PrintTemplates | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PrintTemplateManagementTest.php; tests/Feature/InvoiceManagementTest.php; tests/Feature/POSManagementTest.php; |
| BRD-19-03 / Mandatory | Footer updates / PrintTemplates | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PrintTemplateManagementTest.php; tests/Feature/InvoiceManagementTest.php; tests/Feature/POSManagementTest.php; |
| BRD-19-04 / Mandatory | Paper-size validation / PrintTemplates | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PrintTemplateManagementTest.php; tests/Feature/InvoiceManagementTest.php; tests/Feature/POSManagementTest.php; |
| BRD-19-05 / Mandatory | Duplicate marker history / PrintTemplates | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PrintTemplateManagementTest.php; tests/Feature/InvoiceManagementTest.php; tests/Feature/POSManagementTest.php; |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-19-06 / Mandatory | Quotation PDF / PrintTemplates | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PrintTemplateManagementTest.php; tests/Feature/InvoiceManagementTest.php; tests/Feature/POSManagementTest.php; |
| BRD-19-07 / Mandatory | Invoice PDF / PrintTemplates | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PrintTemplateManagementTest.php; tests/Feature/InvoiceManagementTest.php; tests/Feature/POSManagementTest.php; |
| BRD-19-08 / Mandatory | POS thermal PDF / PrintTemplates | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PrintTemplateManagementTest.php; tests/Feature/InvoiceManagementTest.php; tests/Feature/POSManagementTest.php; |
| BRD-19-09 / Mandatory | POS A4 PDF / PrintTemplates | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PrintTemplateManagementTest.php; tests/Feature/InvoiceManagementTest.php; tests/Feature/POSManagementTest.php; |
| BRD-19-10 / Mandatory | POS A5 PDF / PrintTemplates | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PrintTemplateManagementTest.php; tests/Feature/InvoiceManagementTest.php; tests/Feature/POSManagementTest.php; |
| BRD-19-11 / Mandatory | Customer statement PDF / PrintTemplates | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PrintTemplateManagementTest.php; tests/Feature/InvoiceManagementTest.php; tests/Feature/POSManagementTest.php; |
| BRD-19-12 / Mandatory | Payment receipt output / PrintTemplates | Implemented and Verified; Passed within cited suite; see test inventory for individual assertions | tests/Feature/PrintTemplateManagementTest.php; tests/Feature/InvoiceManagementTest.php; tests/Feature/POSManagementTest.php; |
| BRD-19-13 / Mandatory | Every branding and title setting applied consistently / PrintTemplates | Partially Implemented; Not verified end to end | tests/Feature/PrintTemplateManagementTest.php; tests/Feature/InvoiceManagementTest.php; tests/Feature/POSManagementTest.php; PDF/font/title/paper/history gaps have corrections and fixture evidence. Complete physical signature/page-break/layout acceptance still needs owner hardware testing. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-19-14 / Mandatory | A5 receipt PDF download / PrintTemplates | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up workflows preserve existing ledgers, original records and history. See pending-completion-report.md for bounded return/costing limitations. |
| BRD-19-15 / Mandatory | A6 card PDF download / PrintTemplates | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up workflows preserve existing ledgers, original records and history. See pending-completion-report.md for bounded return/costing limitations. |
| BRD-19-16 / Mandatory | Supplier document printing / PrintTemplates | Implemented and Verified; Passed in final disposable MySQL verification | tests/Feature/StockPurchasingWorkflowTest.php; tests/Feature/PrintGapCompletionTest.php; scripts/verify-pending-concurrency.php; Follow-up workflows preserve existing ledgers, original records and history. See pending-completion-report.md for bounded return/costing limitations. |
| BRD-19-17 / Mandatory | Physical printer output / PrintTemplates | Blocked; Not verified end to end | tests/Feature/PrintTemplateManagementTest.php; tests/Feature/InvoiceManagementTest.php; tests/Feature/POSManagementTest.php; No physical printer acceptance was performed. |
| BRD-19-18 / Mandatory | All signature and page break acceptance / PrintTemplates | Partially Implemented; Not verified end to end | tests/Feature/PrintTemplateManagementTest.php; tests/Feature/InvoiceManagementTest.php; tests/Feature/POSManagementTest.php; PDF/font/title/paper/history gaps have corrections and fixture evidence. Complete physical signature/page-break/layout acceptance still needs owner hardware testing. |

## BRD section 20

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-20-01 / Mandatory | Complete thirteen-stage repair workflow / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Core operational workflows and selected cross-module scenarios now pass. Full owner/hardware/configuration acceptance remains pending. |
| BRD-20-02 / Mandatory | Independent POS workflow / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Core operational workflows and selected cross-module scenarios now pass. Full owner/hardware/configuration acceptance remains pending. |
| BRD-20-03 / Mandatory | Supplier purchase workflow / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Core operational workflows and selected cross-module scenarios now pass. Full owner/hardware/configuration acceptance remains pending. |

## BRD section 21

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-21-01 / Optional | Warranty tracking / System | Implemented but Not Tested; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Recommended optional extensions, not silently added as mandatory scope. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-21-02 / Optional | Repeat repair tracking / System | Implemented but Not Tested; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Recommended optional extensions, not silently added as mandatory scope. |
| BRD-21-03 / Optional | SMS or WhatsApp API / System | Implemented but Not Tested; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Recommended optional extensions, not silently added as mandatory scope. |
| BRD-21-04 / Optional | Ready-for-collection notification / System | Implemented but Not Tested; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Recommended optional extensions, not silently added as mandatory scope. |
| BRD-21-05 / Optional | Overdue job alerts / System | Implemented but Not Tested; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Recommended optional extensions, not silently added as mandatory scope. |
| BRD-21-06 / Optional | Technician workload enhancements / System | Implemented but Not Tested; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Recommended optional extensions, not silently added as mandatory scope. |
| BRD-21-07 / Optional | Low stock notification enhancements / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Core reports/history/settings exist; proactive notifications not verified. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-21-08 / Optional | Approval history enhancements / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Core reports/history/settings exist; proactive notifications not verified. |
| BRD-21-09 / Optional | Custom branding enhancements / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Core reports/history/settings exist; proactive notifications not verified. |

## BRD section 22

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-22-01 / Mandatory | Source code and documentation handover / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Existing module documentation exists; formal handover and completeness review pending. |
| BRD-22-02 / Mandatory | User manual / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Existing module documentation exists; formal handover and completeness review pending. |
| BRD-22-03 / Mandatory | Administrator manual / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Existing module documentation exists; formal handover and completeness review pending. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-22-04 / Mandatory | Setup guidance / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Existing module documentation exists; formal handover and completeness review pending. |
| BRD-22-05 / Mandatory | Training / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Needs owner/vendor participation; no signatures fabricated. |
| BRD-22-06 / Mandatory | Signed UAT / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Needs owner/vendor participation; no signatures fabricated. |
| BRD-22-07 / Mandatory | License and warranty agreement / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Needs owner/vendor participation; no signatures fabricated. |
| BRD-22-08 / Mandatory | Installer or agreed deployment package / System | Not Applicable; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Packaging and deployment expressly excluded from this step. |

# BRD section 23

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-23-01 / Mandatory | Full repair acceptance / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Automated source workflows now pass; business-owner signatures, physical devices and approved operational backup configuration remain pending. |
| BRD-23-02 / Mandatory | Full POS acceptance / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Automated source workflows now pass; business-owner signatures, physical devices and approved operational backup configuration remain pending. |
| BRD-23-03 / Mandatory | Supplier acceptance / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Automated source workflows now pass; business-owner signatures, physical devices and approved operational backup configuration remain pending. |
| BRD-23-04 / Mandatory | Physical printing acceptance / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Automated source workflows now pass; business-owner signatures, physical devices and approved operational backup configuration remain pending. |
| BRD-23-05 / Mandatory | Backup operational acceptance / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Automated source workflows now pass; business-owner signatures, physical devices and approved operational backup configuration remain pending. |

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-23-06 / Mandatory | Concurrent-user acceptance / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Automated source workflows now pass; business-owner signatures, physical devices and approved operational backup configuration remain pending. |
| BRD-23-07 / Mandatory | Written owner approval / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Automated source workflows now pass; business-owner signatures, physical devices and approved operational backup configuration remain pending. |

## BRD section 24

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-24-01 / Mandatory | Agreed software cost / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Commercial agreement not provided; no amounts invented. |
| BRD-24-02 / Mandatory | Third-party recurring charges / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Commercial agreement not provided; no amounts invented. |
| BRD-24-03 / Mandatory | Payment and commercial terms / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Commercial agreement not provided; no amounts invented. |

# BRD section 25

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-25-01 / Mandatory | Warranty duration / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Contractual commitments require agreement outside technical testing. |
| BRD-25-02 / Mandatory | Support hours / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Contractual commitments require agreement outside technical testing. |
| BRD-25-03 / Mandatory | Remote support agreement / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Contractual commitments require agreement outside technical testing. |
| BRD-25-04 / Mandatory | Support fees / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Contractual commitments require agreement outside technical testing. |
| BRD-25-05 / Mandatory | Recovery assistance / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Contractual commitments require agreement outside technical testing. |
| BRD-25-06 / Mandatory | Future updates and migration agreement / System | Blocked; Not executed / unavailable | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Contractual commitments require agreement outside technical testing. |

# BRD section 26

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-26-01 / Mandatory | Business data ownership / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Source and backup mechanisms exist; complete handover/contract evidence unavailable; no operational export performed. |
| BRD-26-02 / Mandatory | Recoverable data export / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Source and backup mechanisms exist; complete handover/contract evidence unavailable; no operational export performed. |
| BRD-26-03 / Mandatory | Maintainable source and documentation / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Source and backup mechanisms exist; complete handover/contract evidence unavailable; no operational export performed. |
| BRD-26-04 / Mandatory | Confidentiality / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Source and backup mechanisms exist; complete handover/contract evidence unavailable; no operational export performed. |
| BRD-26-05 / Mandatory | License terms and fee disclosure / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Source and backup mechanisms exist; complete handover/contract evidence unavailable; no operational export performed. |

## BRD section 27

| ID / scope | Requirement / module | Status / result | Evidence / boundary |
|---|---|---|---|
| BRD-27-01 / Mandatory | Complete integrated business solution / System | Partially Implemented; Not verified end to end | tests/Feature/SystemSupportTest.php; tests/Feature/Auth/AuthenticationTest.php; tests/Unit/AuditSafetyTest.php; Core operational workflows and selected cross-module scenarios now pass. Full owner/hardware/configuration acceptance remains pending. |

# 25. Evidence index

| Evidence | Purpose |
|---|---|
| step13-tests.log / step13-junit.xml | Raw fresh console and exact case/assertion results |
| implementation.json | Redacted route/role/version/backup/session metadata |
| schema-before.json / schema-after.json / database-summary.json | Operational read-only schema/FK/index/count comparison |
| source-inventory.json | All inspected source file hashes, declarations and line counts |
| brd-text.txt / original BRD PDF | Original requirement source; extracted text is not a translation |
| screenshots.json / screenshot-capture.log / screenshot-server.log | Actual capture manifest and isolated session logs |
| sources/feature-catalogue.json / test-cases.json / defects.json | Editable machine-readable catalogues generated from evidence |
| sources/*relationships.svg / evidence/*.relationships.json | Rendered diagrams and exact verified FK edge lists |
| docs/testing/brd-compliance-matrix.md | Original complete traceability matrix |
| document-verification.json / completion-report.md | Final PDF checks, counts, hashes and remaining limitations |